Hotel Reviews Data Analysis

[GitHub - Prajwalrameshr/sankalp: report using r commands · GitHub]

Team Information

Team Number: 30
Team Name: SANKALP
Section: C

Team Members

- **PRAJWAL R** – 1NT23IS158
- **NAAGA SUMUKH B S** – 1NT23IS136

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.0     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.2     ✓ tibble    3.3.1
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.1
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
df1 <- read_csv("7282_1.csv")
```

```
Rows: 35912 Columns: 19
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (12): address, categories, city, country, name, postalCode, province, r...
dbl   (3): latitude, longitude, reviews.rating
lgl   (2): reviews.doRecommend, reviews.id
dttm  (2): reviews.date, reviews.dateAdded

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df2 <- read_csv("Datafiniti_Hotel_Reviews.csv")
```

```
Rows: 10000 Columns: 25
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (19): id, address, categories, primaryCategories, city, country, keys, ...
dbl   (3): latitude, longitude, reviews.rating
dttm  (3): dateAdded, dateUpdated, reviews.date

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df3 <- read_csv("Datafiniti_Hotel_Reviews_Jun19.csv")
```

```
Rows: 10000 Columns: 26
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (19): id, address, categories, primaryCategories, city, country, keys, ...
dbl   (3): latitude, longitude, reviews.rating
lgl   (1): reviews.dateAdded
dttm  (3): dateAdded, dateUpdated, reviews.date

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df <- bind_rows(df1, df2, df3)
```

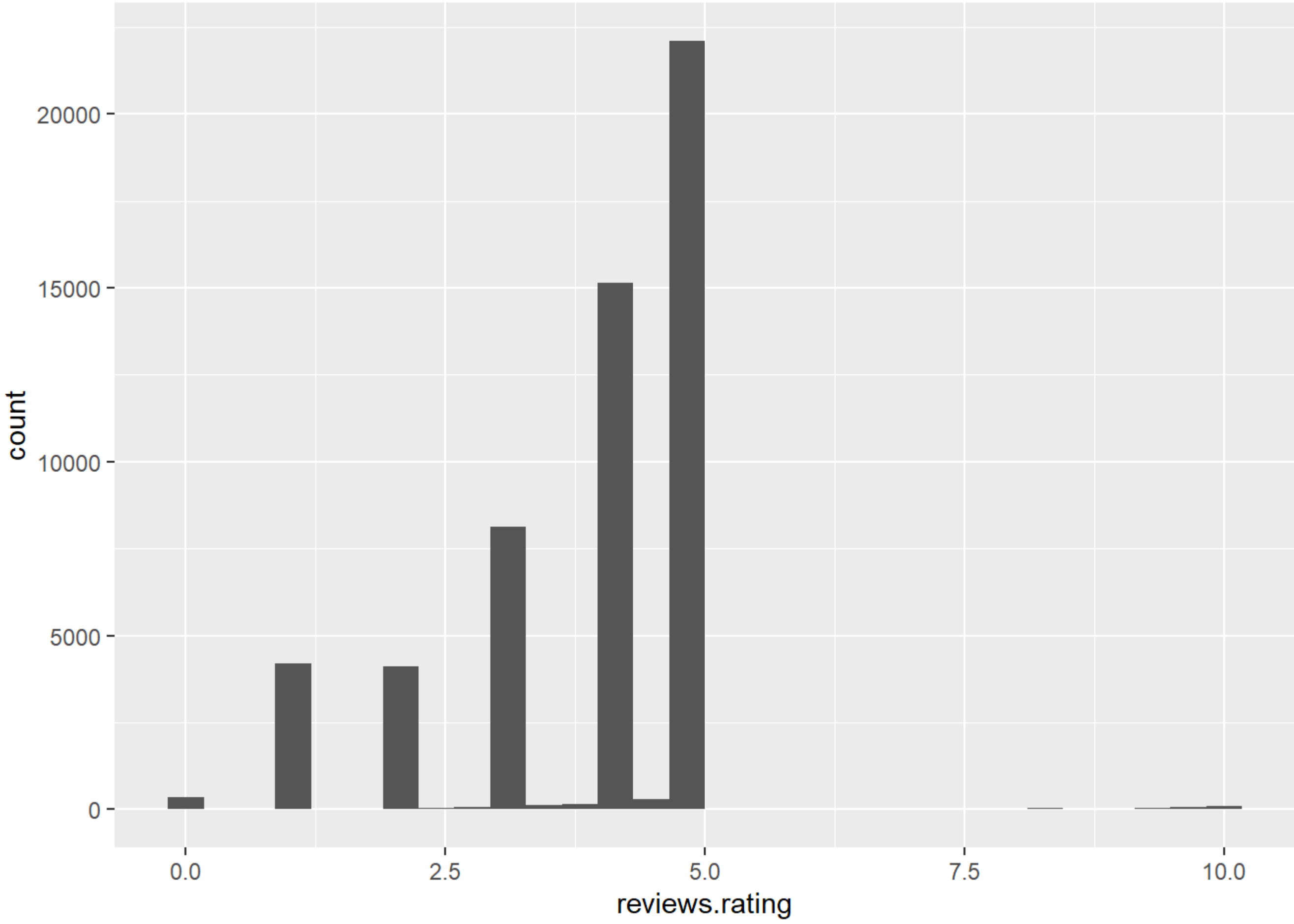
```r
head(df)
```

```
# A tibble: 6 × 28
  address  categories city  country latitude longitude name  postalCode province
  <chr>    <chr>      <chr> <chr>      <dbl>     <dbl> <chr> <chr>      <chr>   
1 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
2 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
3 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
4 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
5 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
6 Riviera… Hotels     Mabl… US          45.4      12.4 Hote… 30126      GA      
# ℹ 19 more variables: reviews.date <dttm>, reviews.dateAdded <dttm>,
#   reviews.doRecommend <lgl>, reviews.id <lgl>, reviews.rating <dbl>,
#   reviews.text <chr>, reviews.title <chr>, reviews.userCity <chr>,
#   reviews.username <chr>, reviews.userProvince <chr>, id <chr>,
#   dateAdded <dttm>, dateUpdated <dttm>, primaryCategories <chr>, keys <chr>,
#   reviews.dateSeen <chr>, reviews.sourceURLs <chr>, sourceURLs <chr>,
#   websites <chr>
```

HISTOGRAM

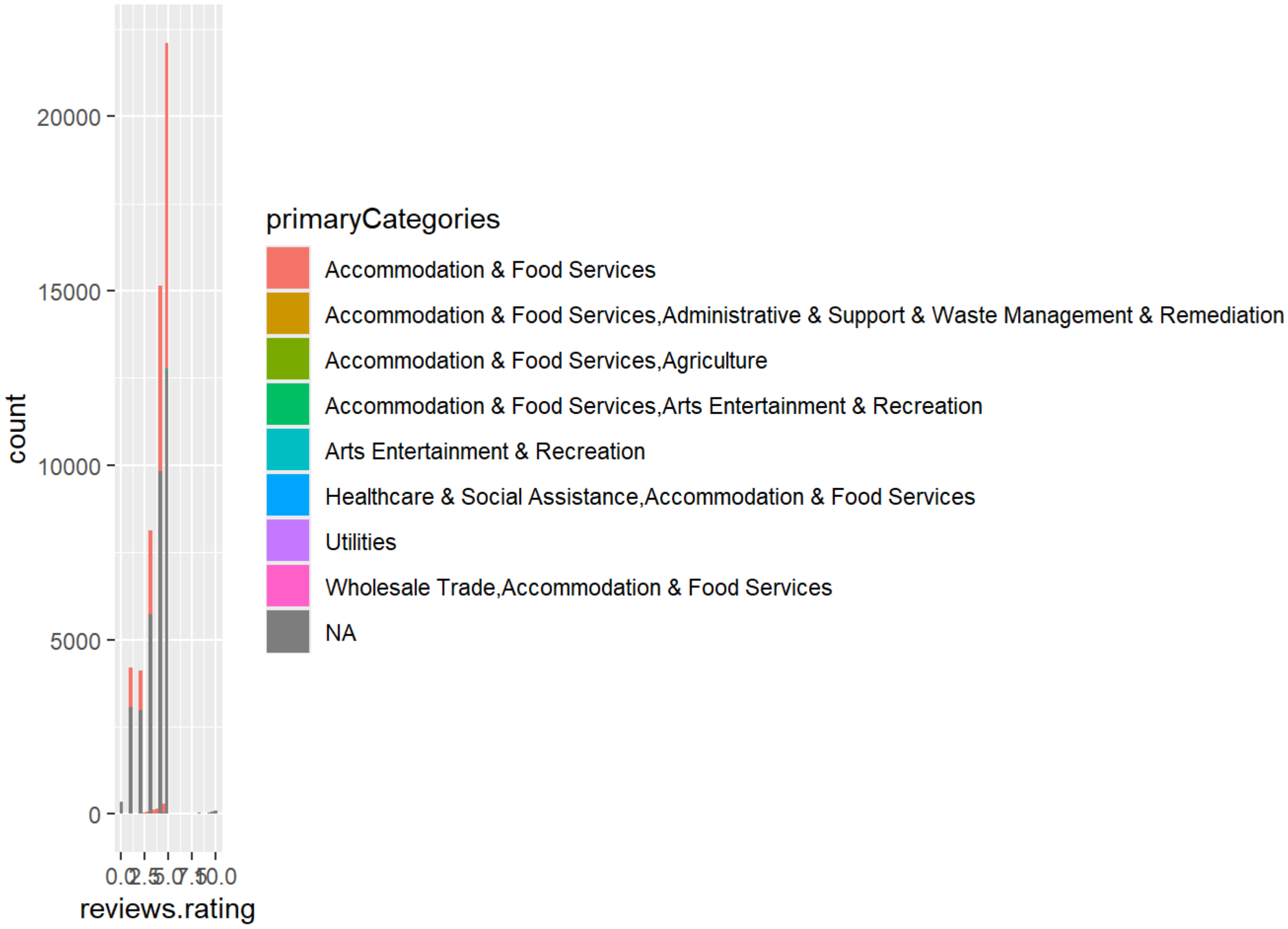
```r
# Simple
ggplot(df, aes(x = reviews.rating)) + geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_bin()`).
```



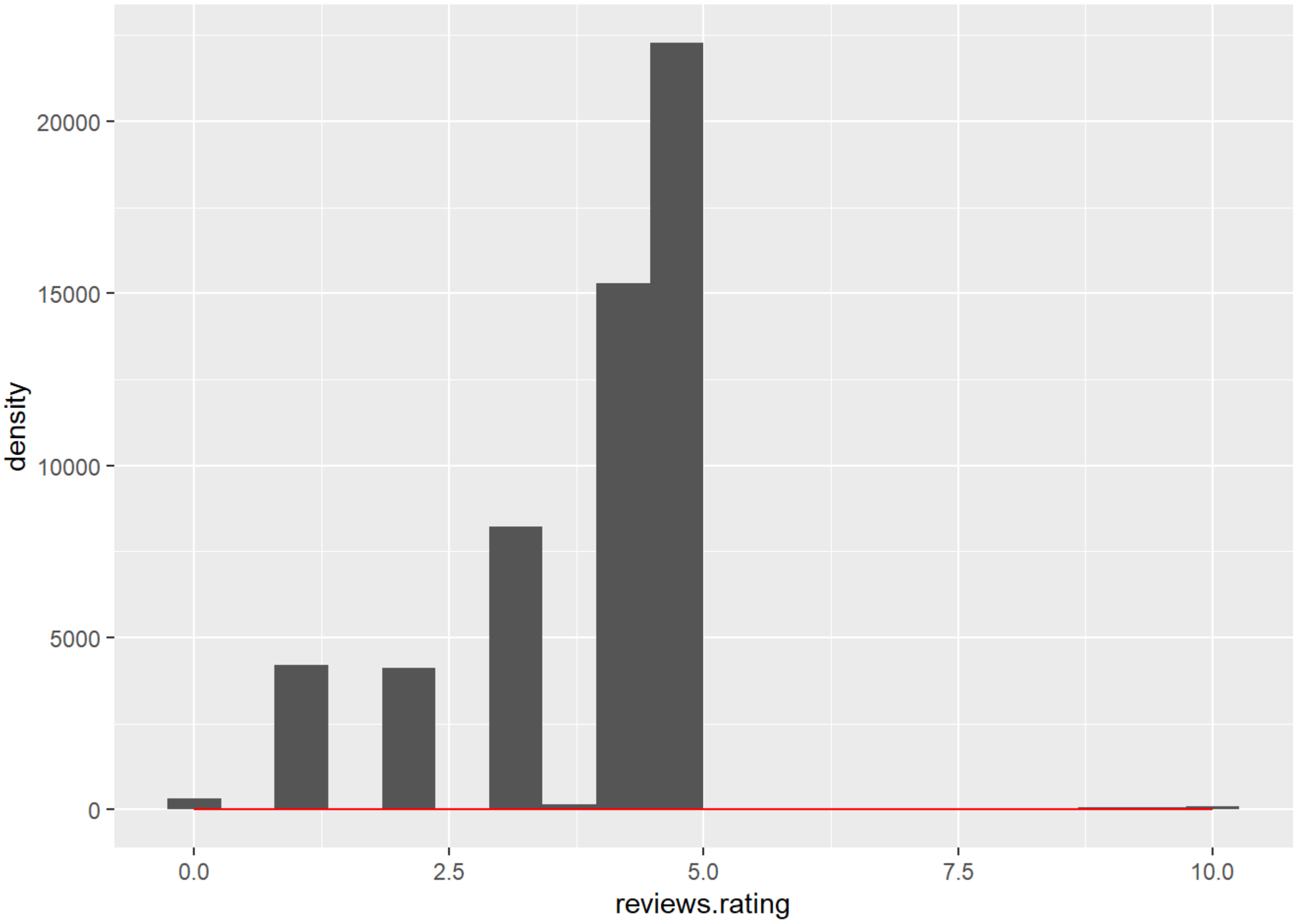

```r
# Moderate
ggplot(df, aes(x = reviews.rating, fill = primaryCategories)) + geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_bin()`).
```



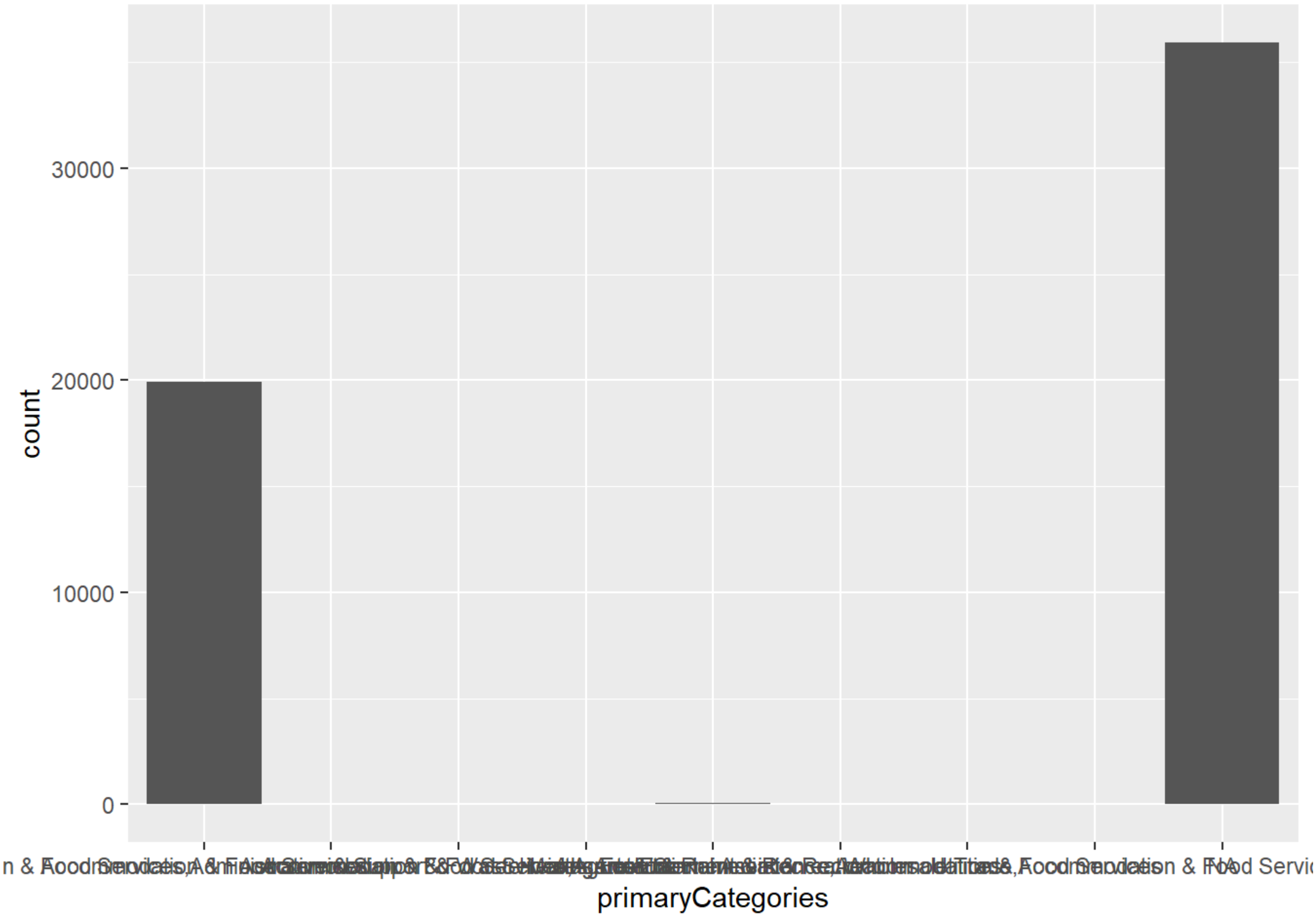

```r
# Advanced
ggplot(df, aes(x = reviews.rating)) +
  geom_histogram(bins = 20) +
  geom_density(color="red")
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_density()`).
```

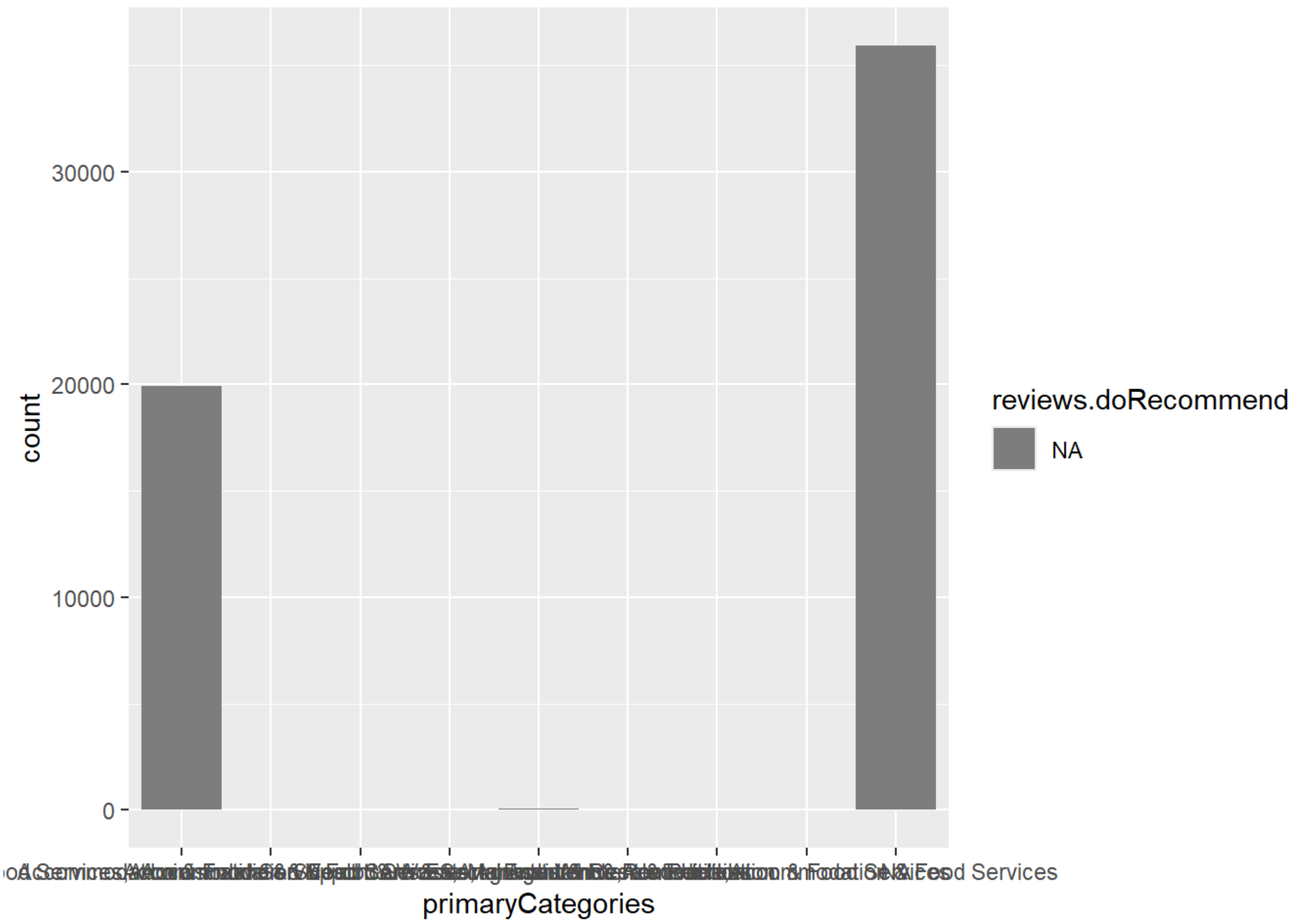


BAR GRAPH

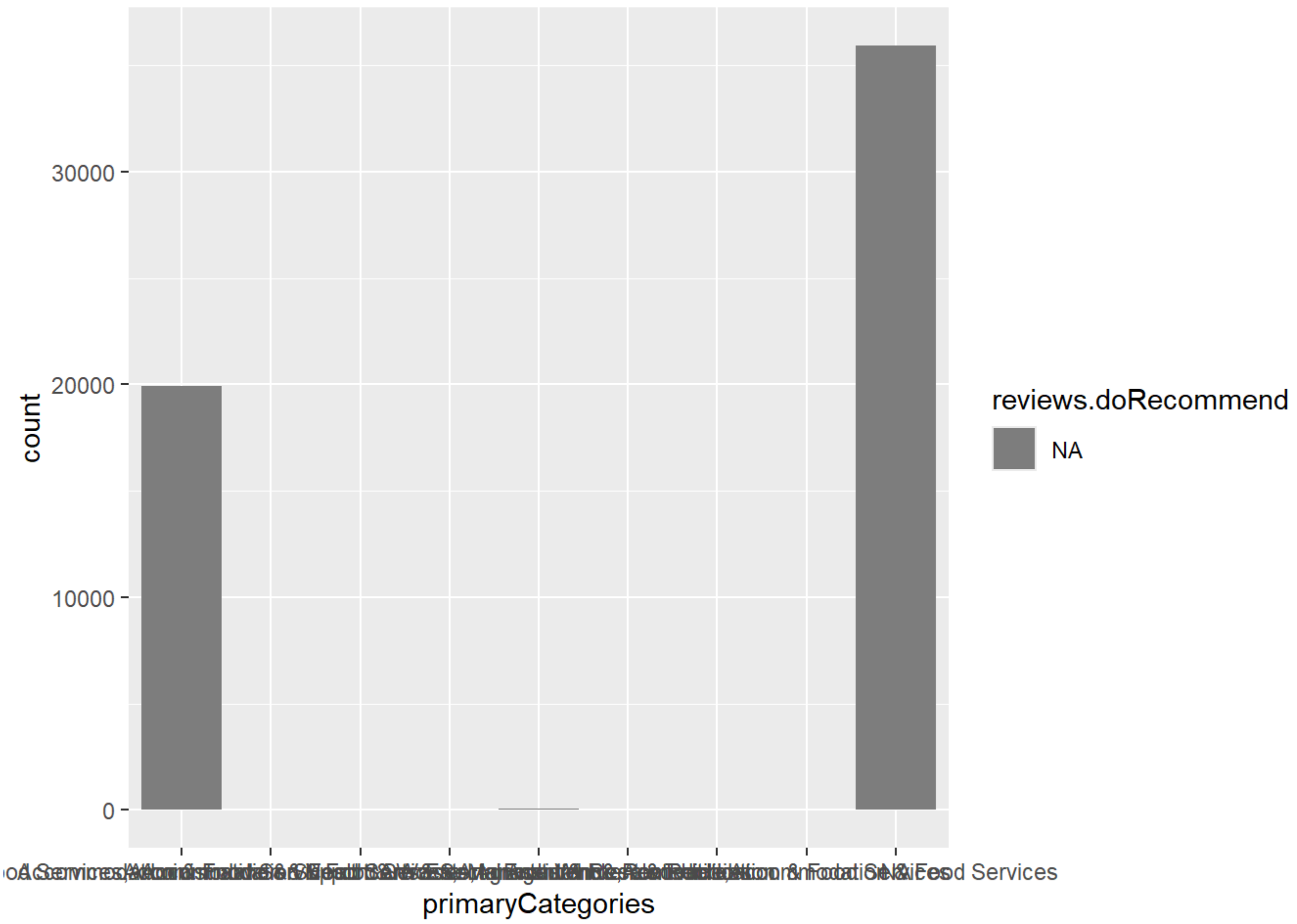
```
# Shows count of each category
ggplot(df, aes(x = primaryCategories)) + geom_bar()
```




```r
# Shows category counts grouped by recommendation
ggplot(df, aes(x = primaryCategories, fill = reviews.doRecommend)) + geom_bar()
```



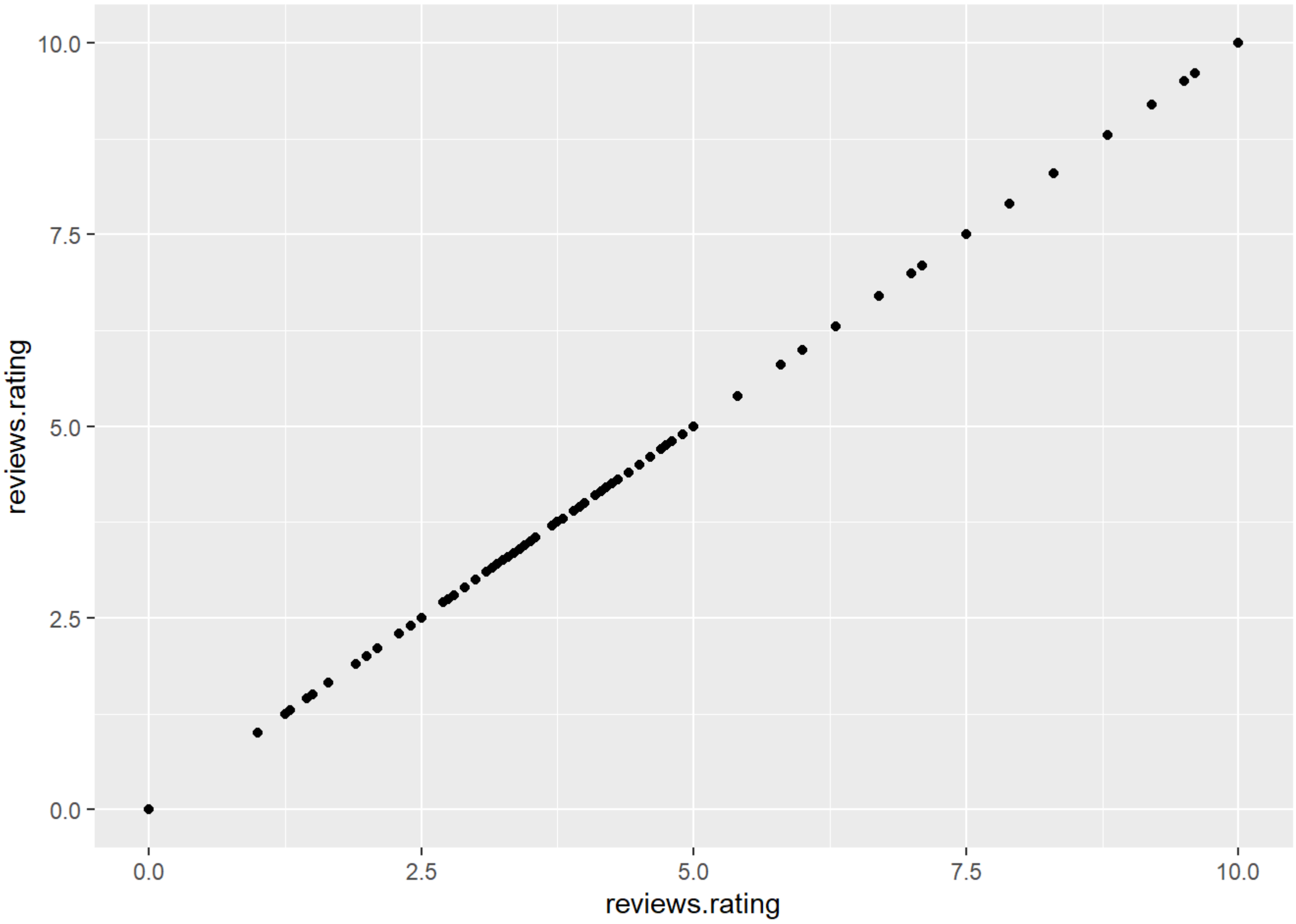

```
# Shows grouped comparison of categories
ggplot(df, aes(x = primaryCategories, fill = reviews.doRecommend)) +
  geom_bar(position = "dodge")
```



SCATTER PLOT

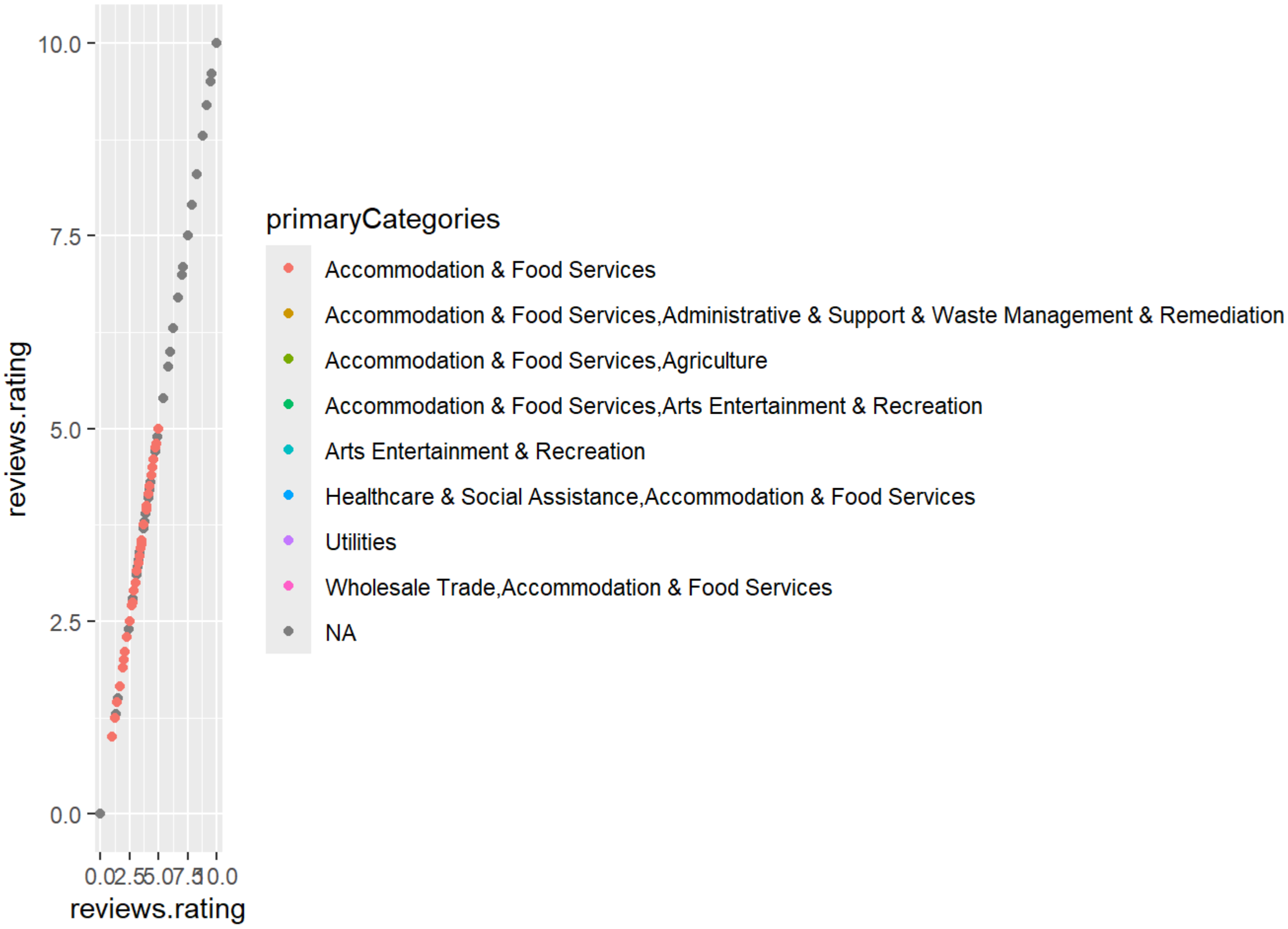
```
# Shows relationship between ratings
ggplot(df, aes(x = reviews.rating, y = reviews.rating)) + geom_point()
```

```
Warning: Removed 862 rows containing missing values or values outside the scale range
(`geom_point()`).
```



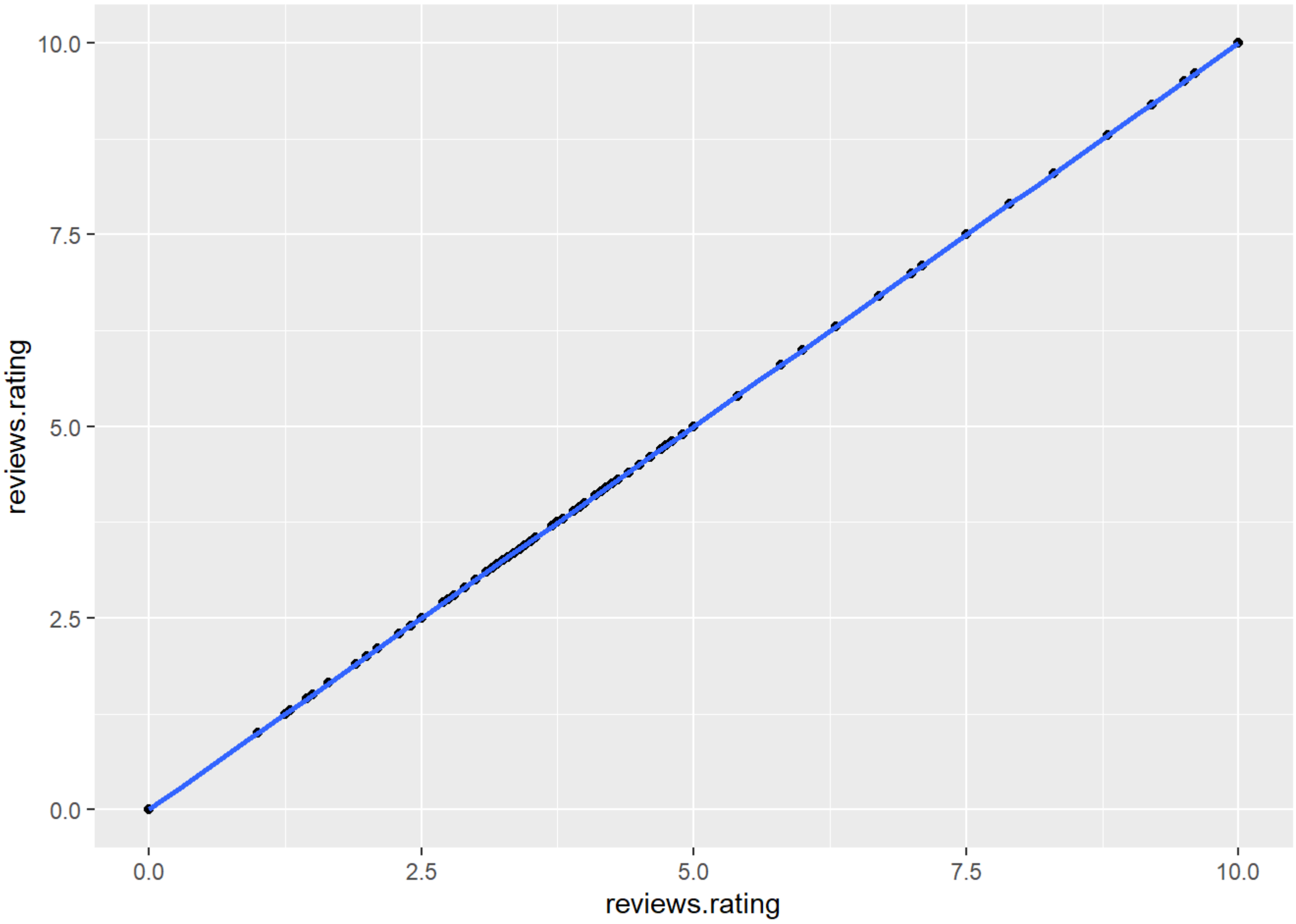

```
# Shows relationship colored by category
ggplot(df, aes(x = reviews.rating, y = reviews.rating, color = primaryCategories)) + geom_poin
```

```
Warning: Removed 862 rows containing missing values or values outside the scale range
(`geom_point()`).
```



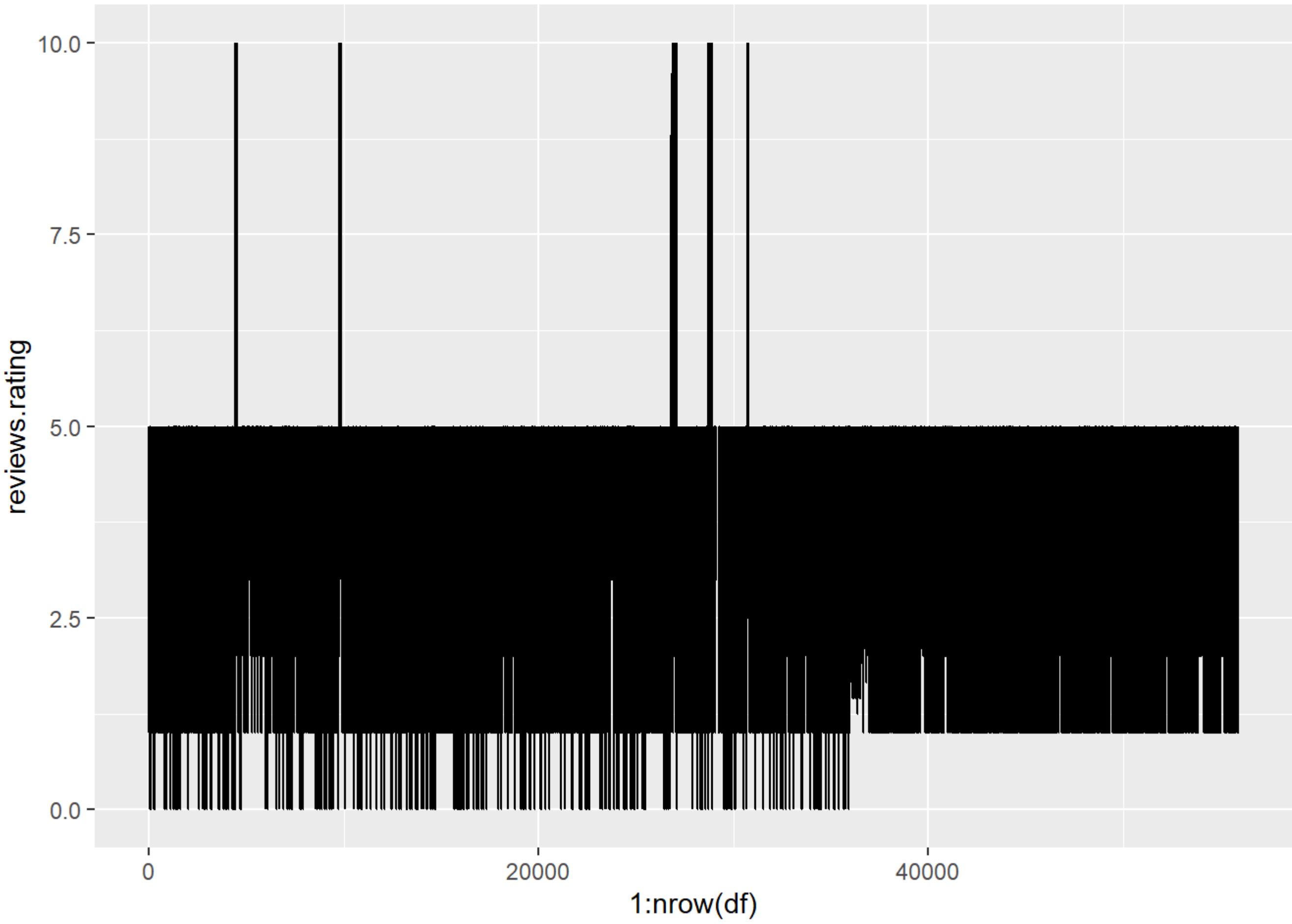

```
# Shows relationship with smooth trend line
ggplot(df, aes(x = reviews.rating, y = reviews.rating)) +
  geom_point() + geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_smooth()`).
Removed 862 rows containing missing values or values outside the scale range
(`geom_point()`).
```

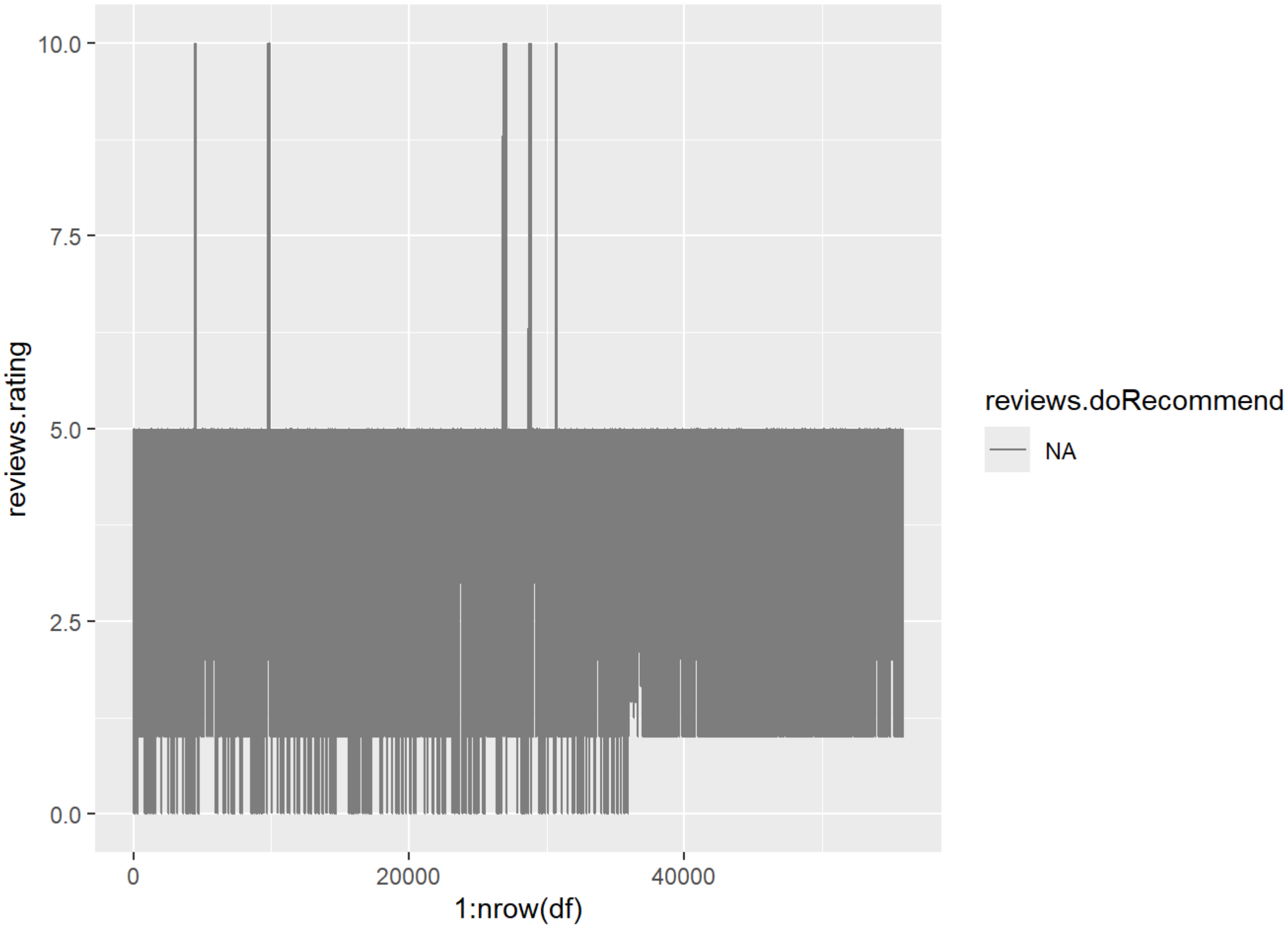


LINE GRAPH

```
# Shows trend of ratings across rows
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating)) + geom_line()
```

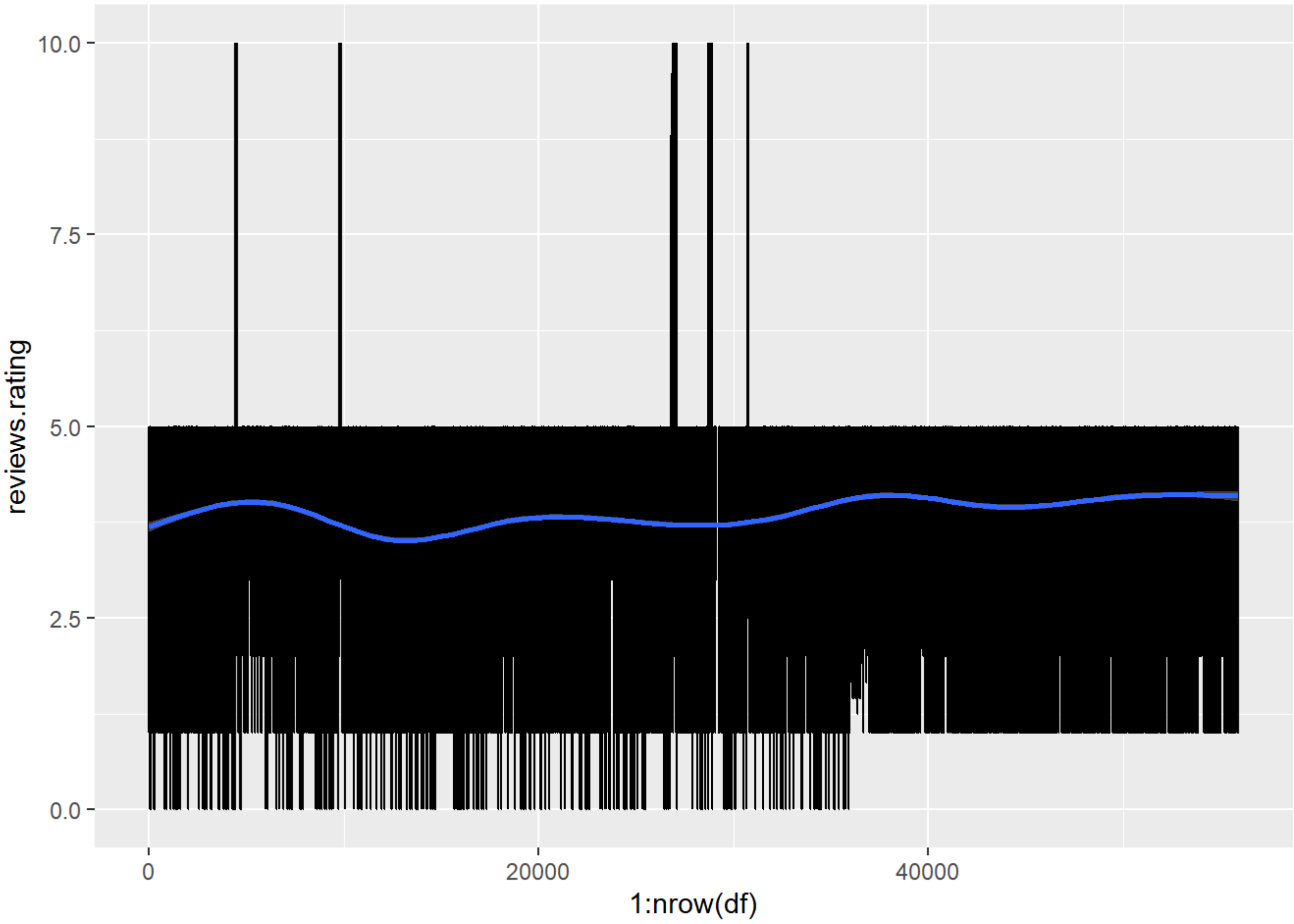



```r
# Shows trend grouped by recommendation
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating, color = reviews.doRecommend)) + geom_line()
```




```
# Shows trend with smoothing line
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating)) +
  geom_line() + geom_smooth()
```

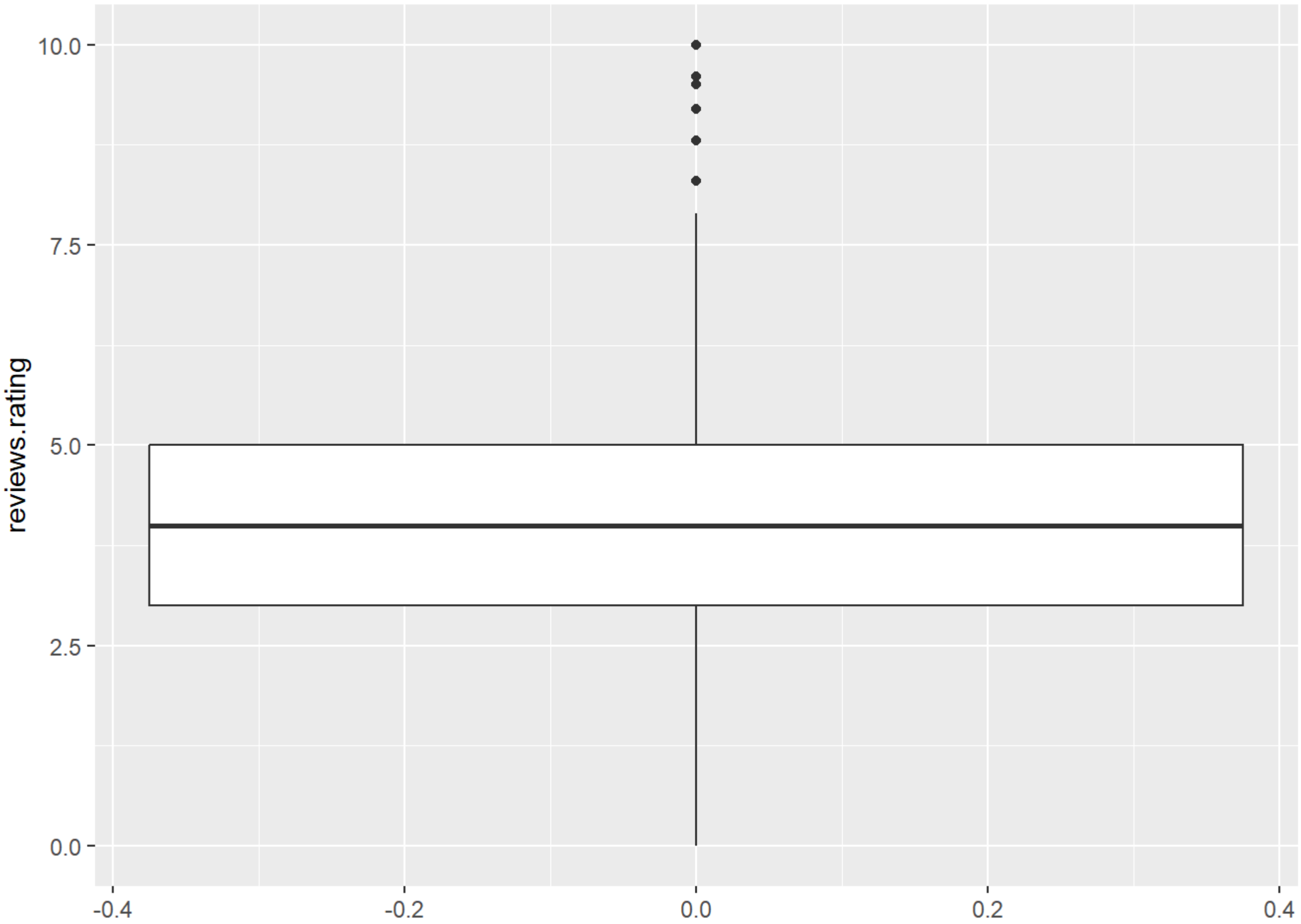
```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_smooth()`).
```



BOX PLOT

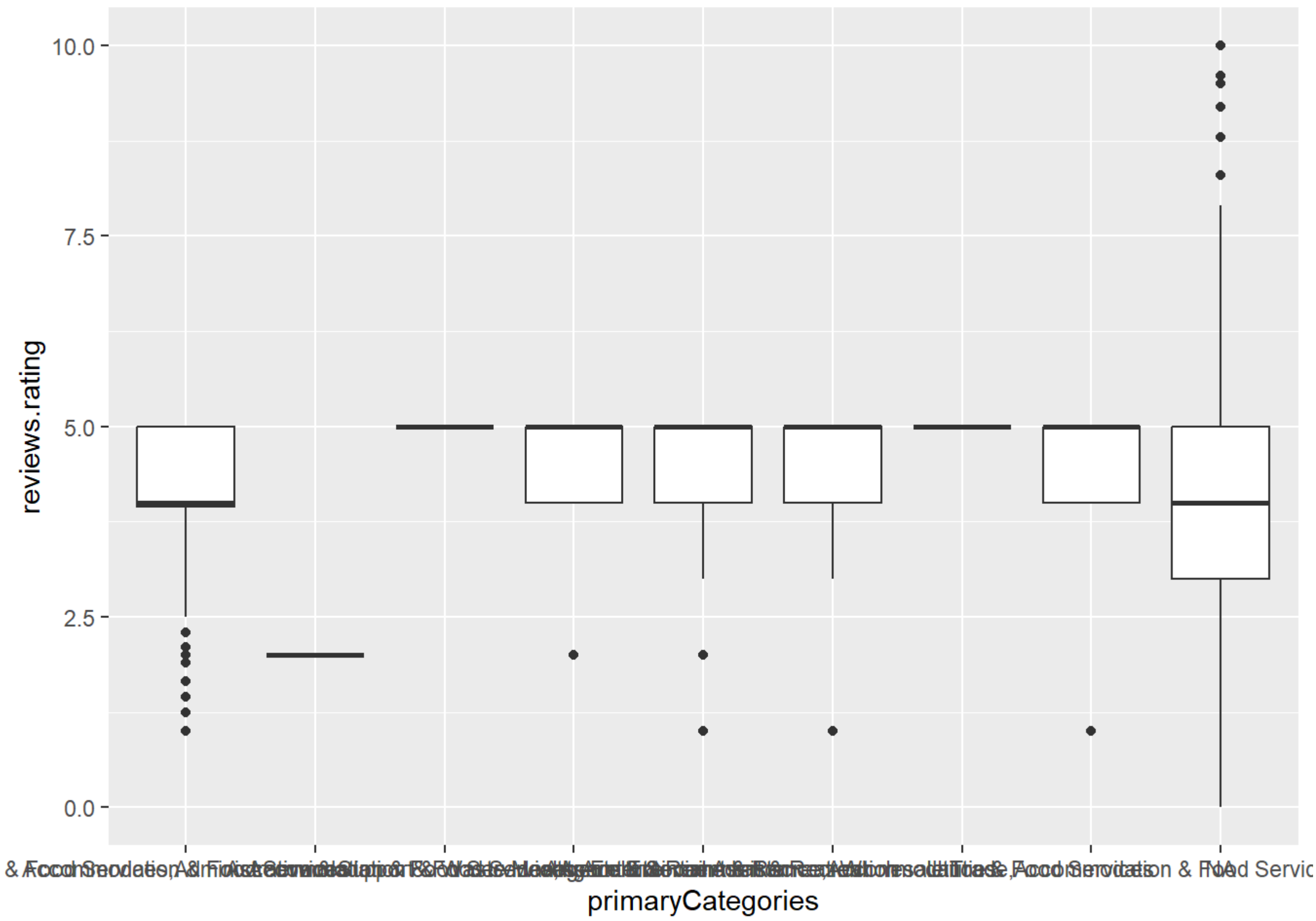
```r
# Shows spread and outliers of ratings
ggplot(df, aes(y = reviews.rating)) + geom_boxplot()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




```r
# Shows rating spread by category
ggplot(df, aes(x = primaryCategories, y = reviews.rating)) + geom_boxplot()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




```r
# Shows boxplot with additional detail using jitter
ggplot(df, aes(x = primaryCategories, y = reviews.rating)) +
  geom_boxplot(notch = TRUE) + geom_jitter()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Notch went outside hinges
ℹ Do you want `notch = FALSE`?
Notch went outside hinges
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Notch went outside hinges
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Notch went outside hinges
ℹ Do you want `notch = FALSE`?

Warning: Removed 862 rows containing missing values or values outside the scale range
(`geom_point()`).
```
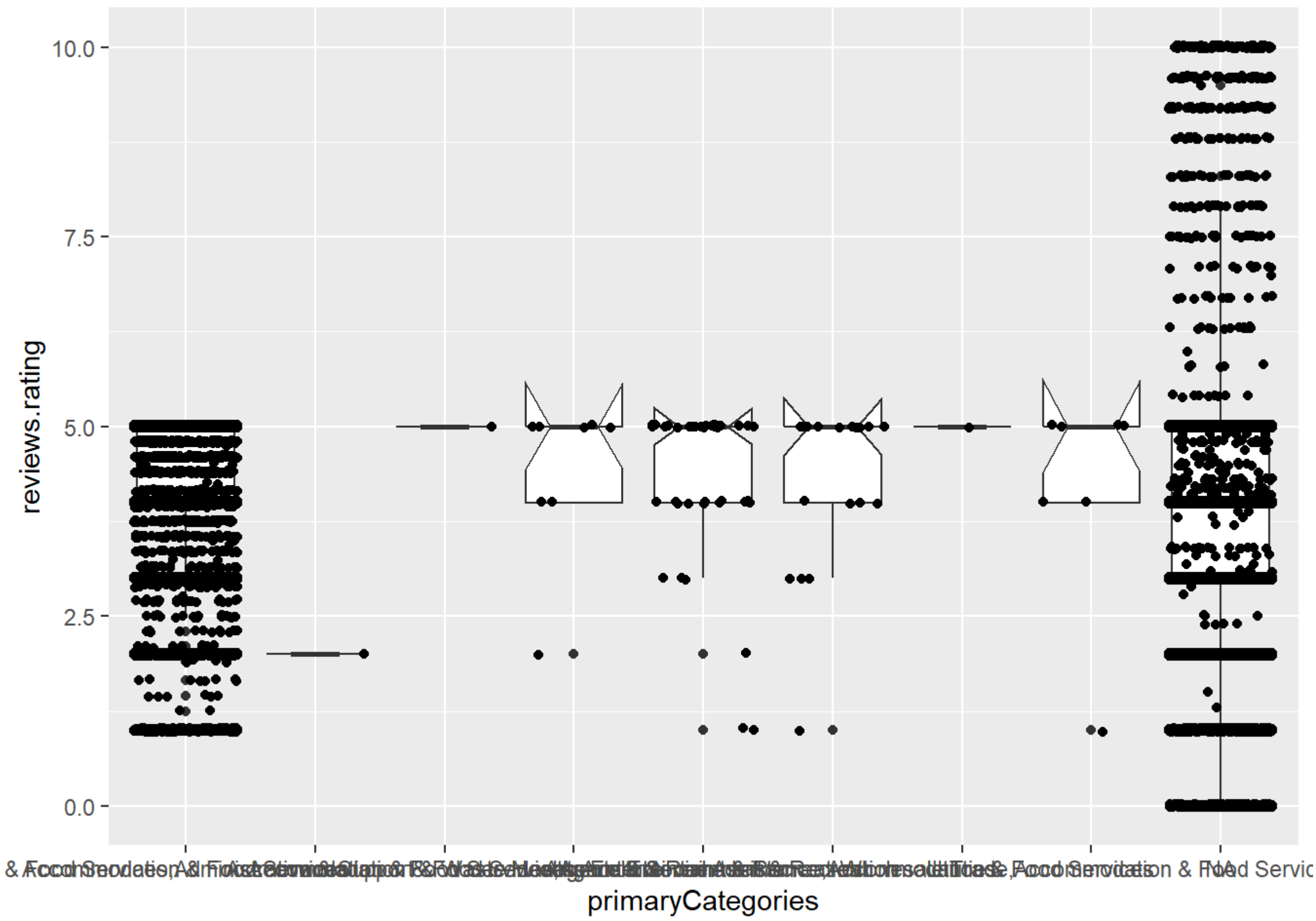
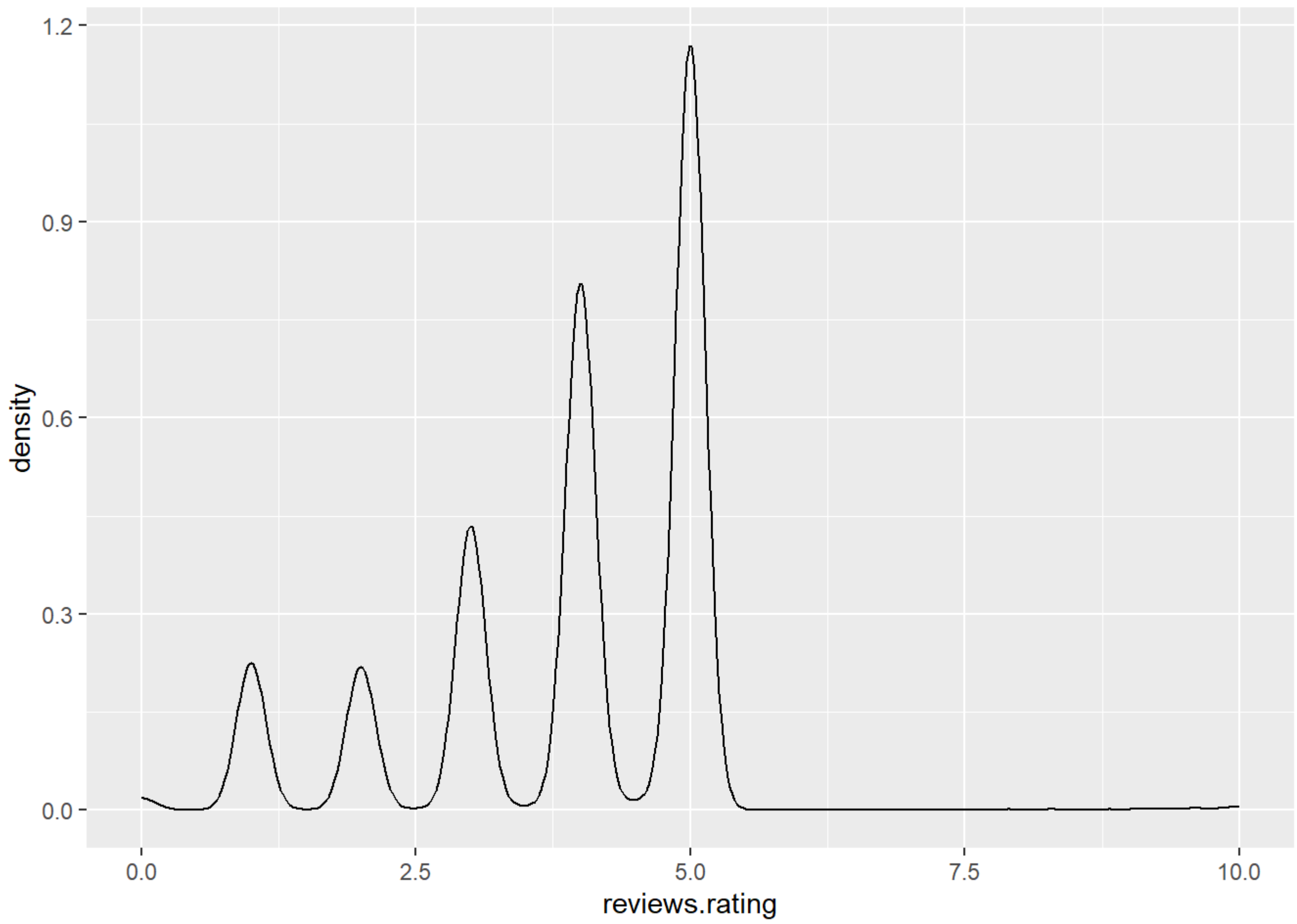



DENSITY PLOT

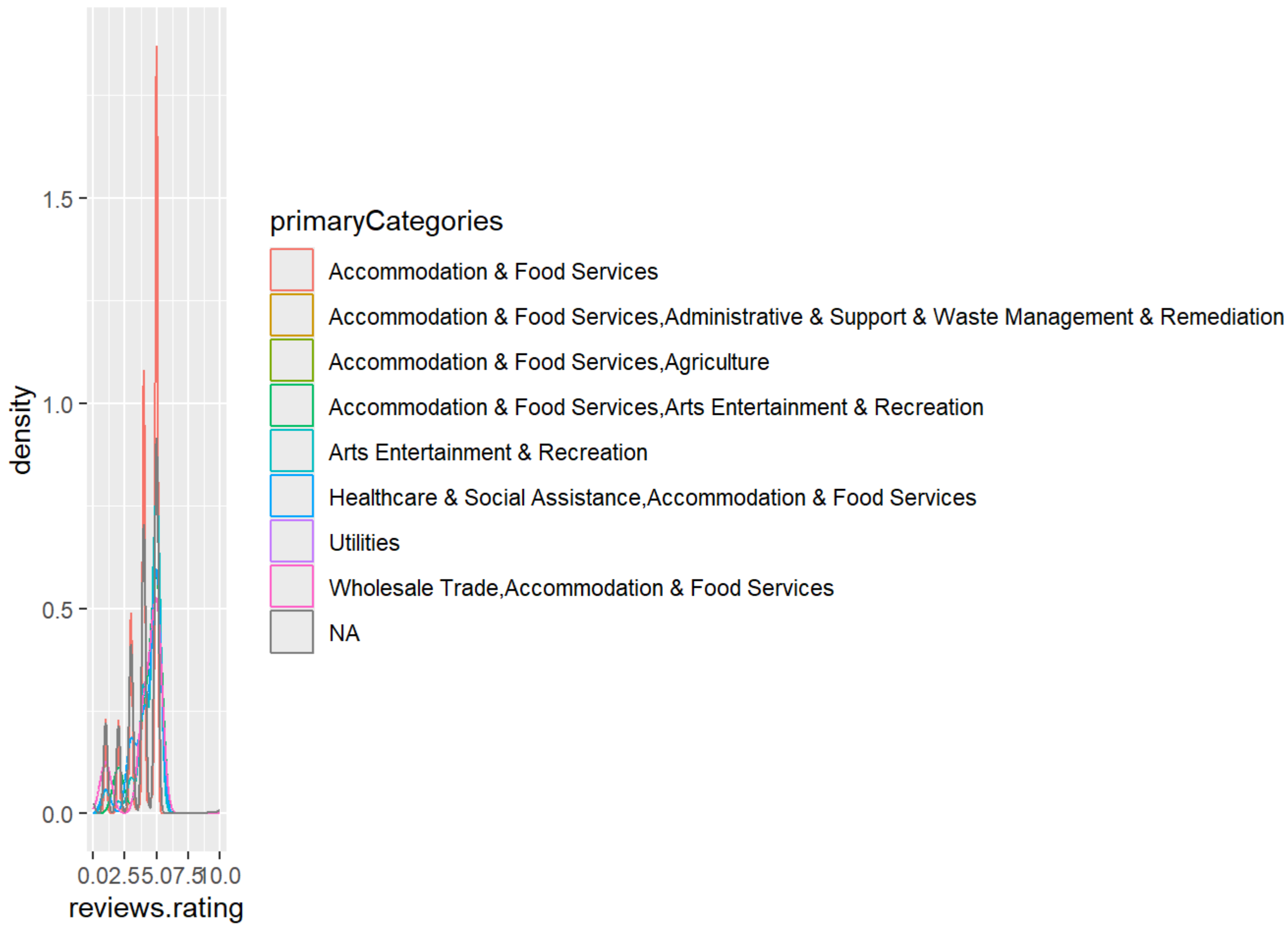
```
# Shows smooth distribution of ratings
ggplot(df, aes(x = reviews.rating)) + geom_density()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_density()`).
```



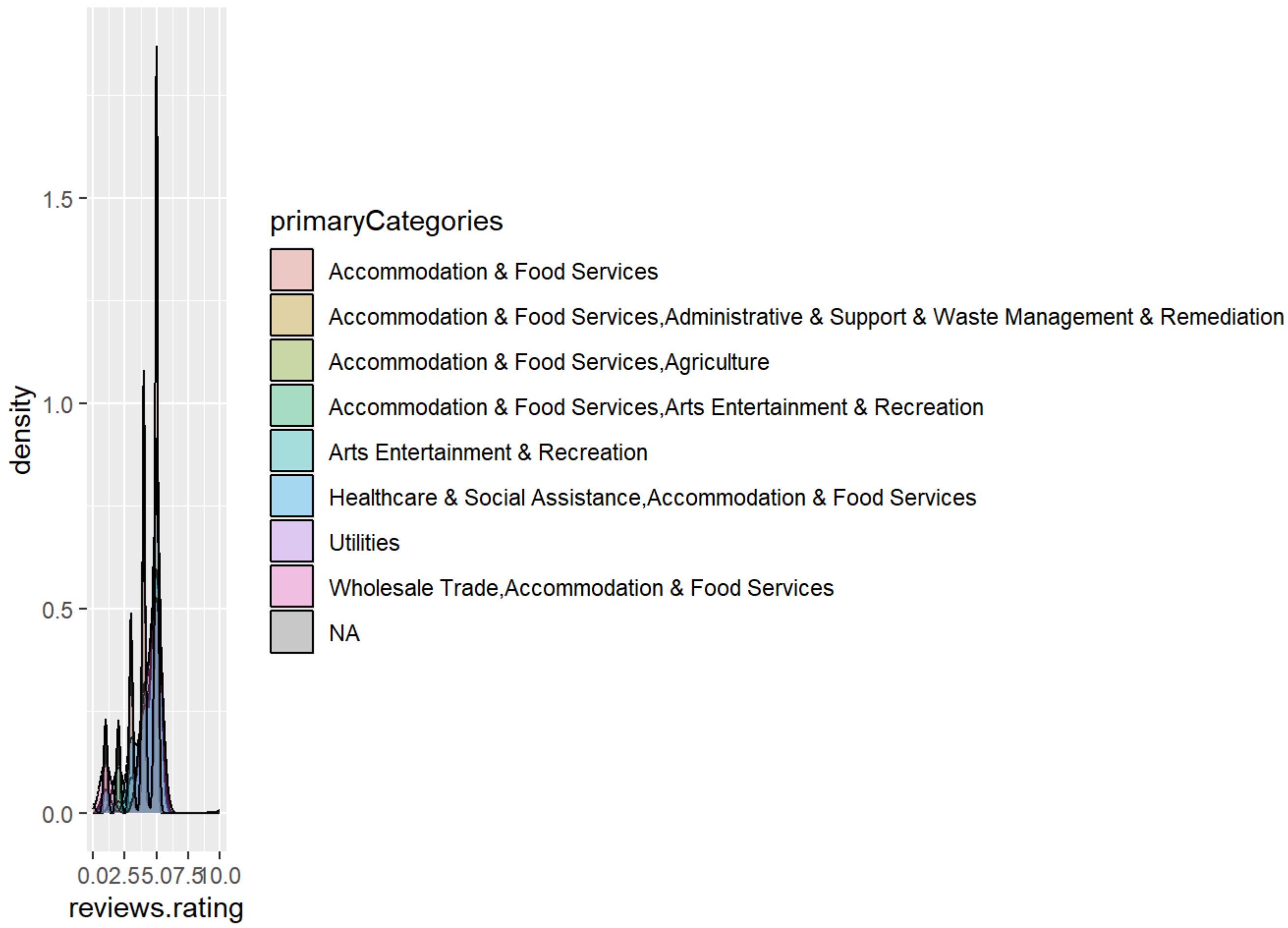

```r
# Shows density grouped by category
ggplot(df, aes(x = reviews.rating, color = primaryCategories)) + geom_density()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.

Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom_density()`).
```




```r
# Shows filled density comparison
ggplot(df, aes(x = reviews.rating, fill = primaryCategories)) +
  geom_density(alpha = 0.3)
```

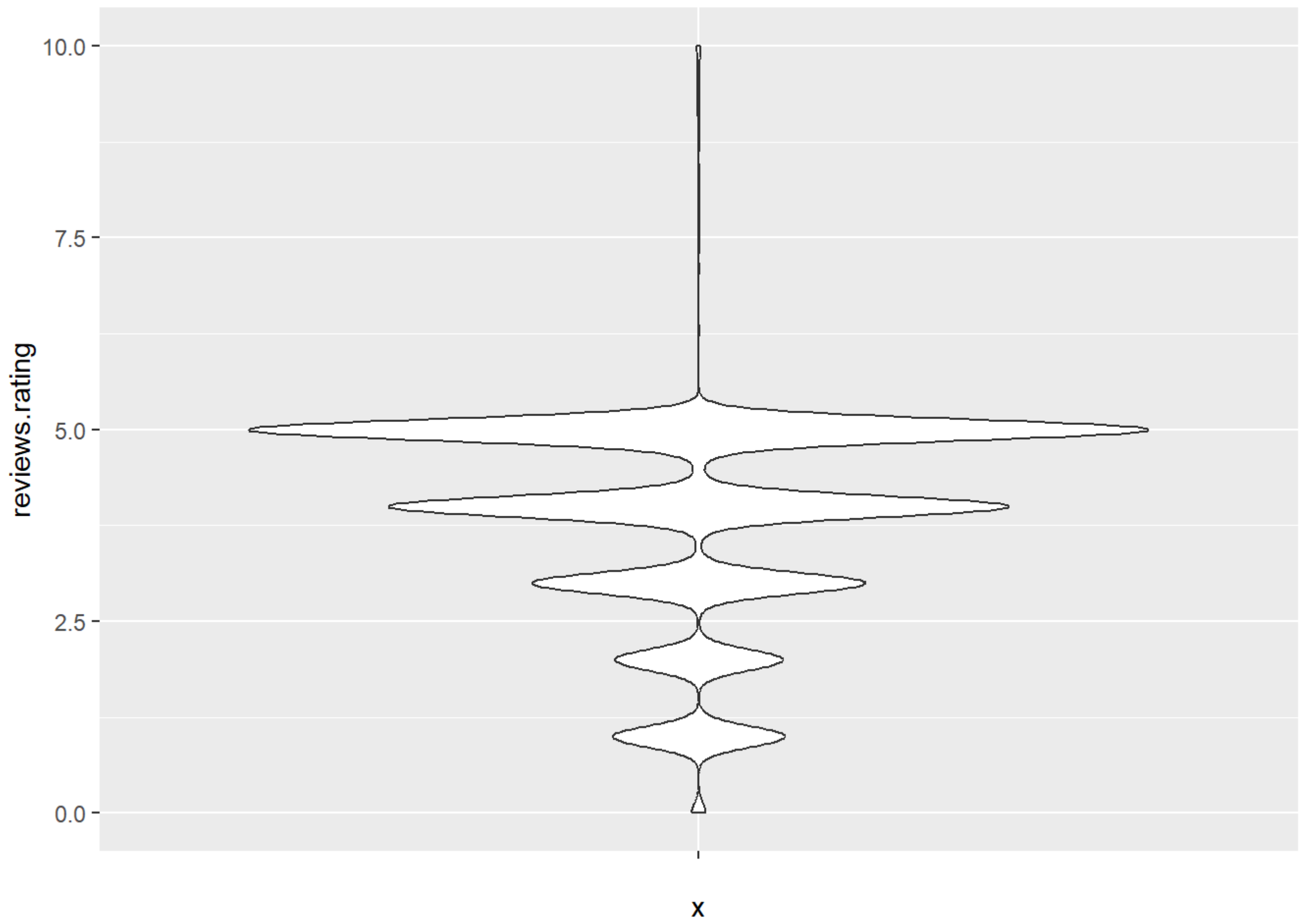
```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.
Groups with fewer than two data points have been dropped.

Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom_density()`).
```



VIOLIN PLOT

```
# Shows distribution using violin plot
ggplot(df, aes(x = "", y = reviews.rating)) + geom_violin()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```




```
# Shows violin plot by category
ggplot(df, aes(x = primaryCategories, y = reviews.rating)) + geom_violin()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_ydensity()`).

Warning: Groups with fewer than two datapoints have been dropped.
ℹ Set `drop = FALSE` to consider such groups for position adjustment purposes.
Groups with fewer than two datapoints have been dropped.
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Groups with fewer than two datapoints have been dropped.
ℹ Set `drop = FALSE` to consider such groups for position adjustment purposes.
```

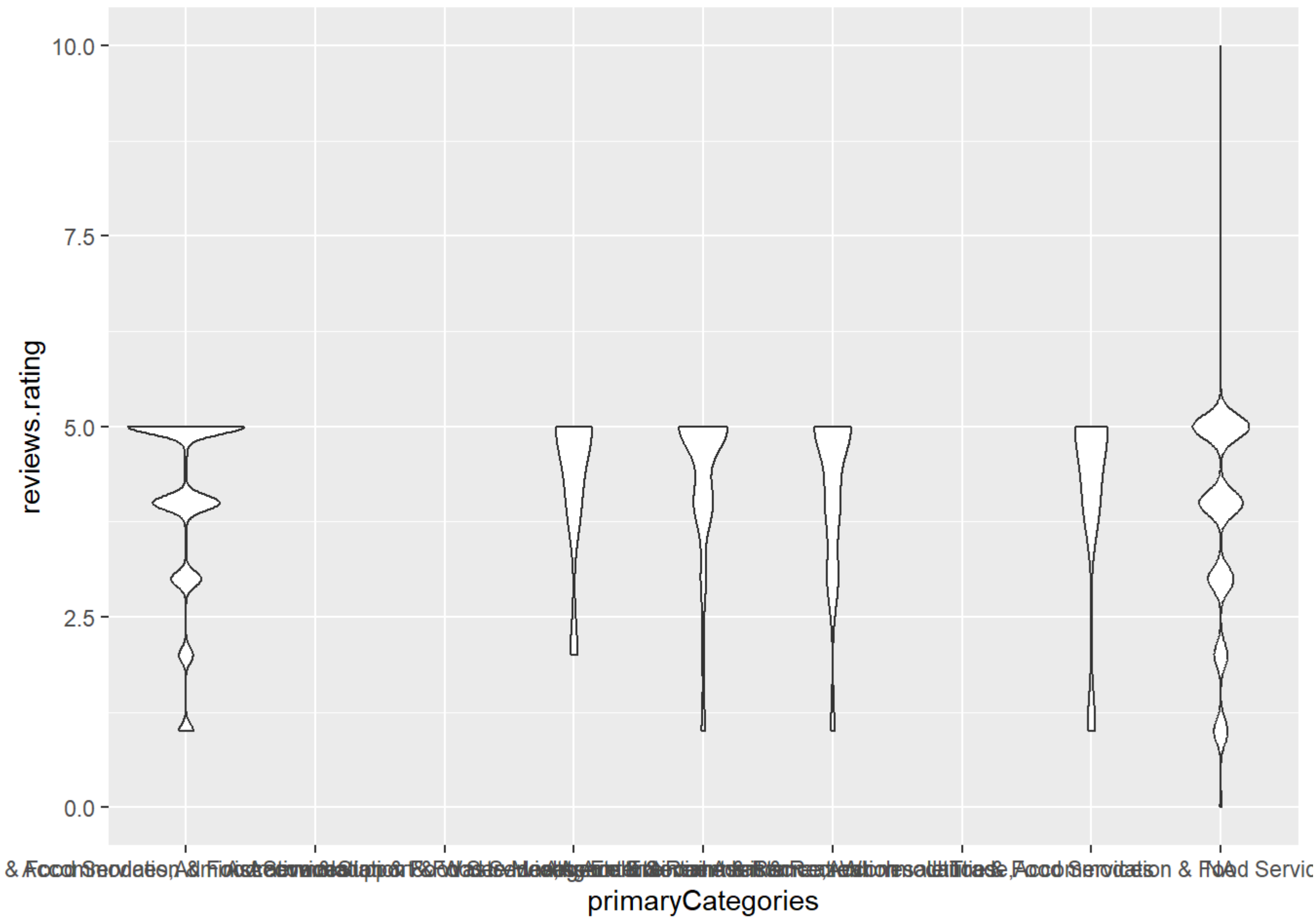


```
# Shows violin with boxplot overlay
ggplot(df, aes(x = primaryCategories, y = reviews.rating)) +
  geom_violin() + geom_boxplot(width = 0.1)
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_ydensity()`).
Groups with fewer than two datapoints have been dropped.
ℹ Set `drop = FALSE` to consider such groups for position adjustment purposes.
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Groups with fewer than two datapoints have been dropped.
ℹ Set `drop = FALSE` to consider such groups for position adjustment purposes.

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```
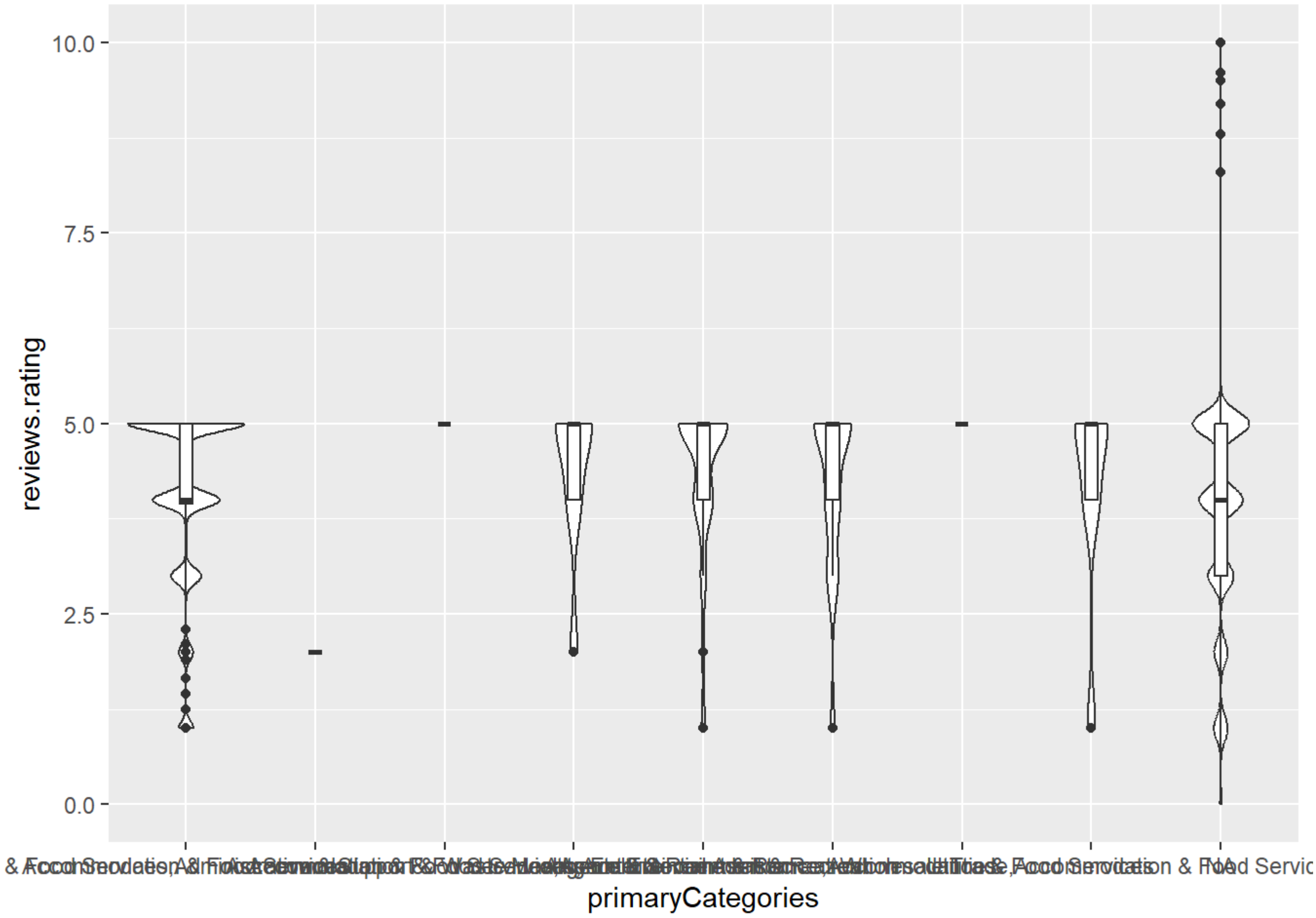
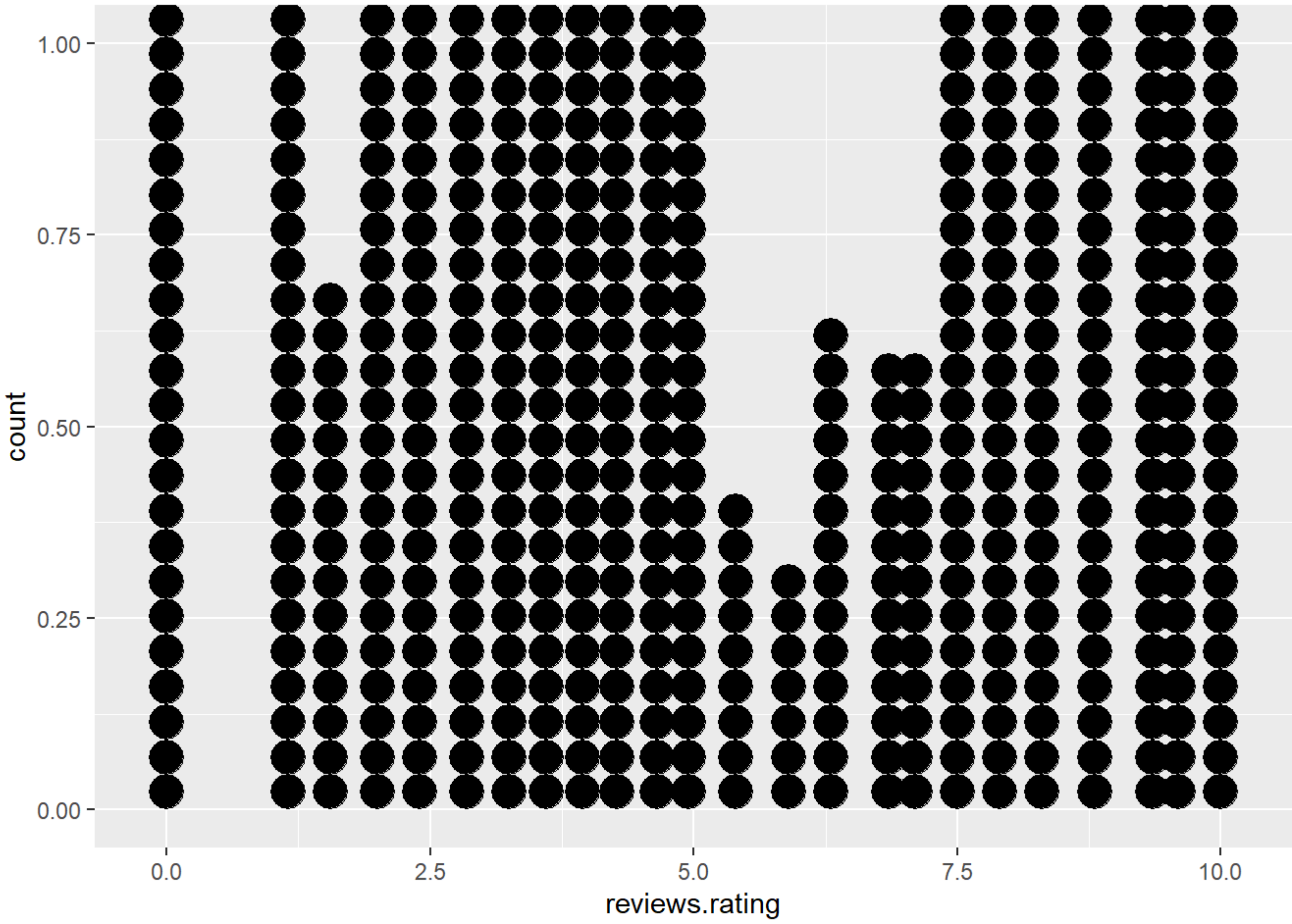



DOT PLOT

```
# Shows distribution using dot plot
ggplot(df, aes(x = reviews.rating)) + geom_dotplot()
```

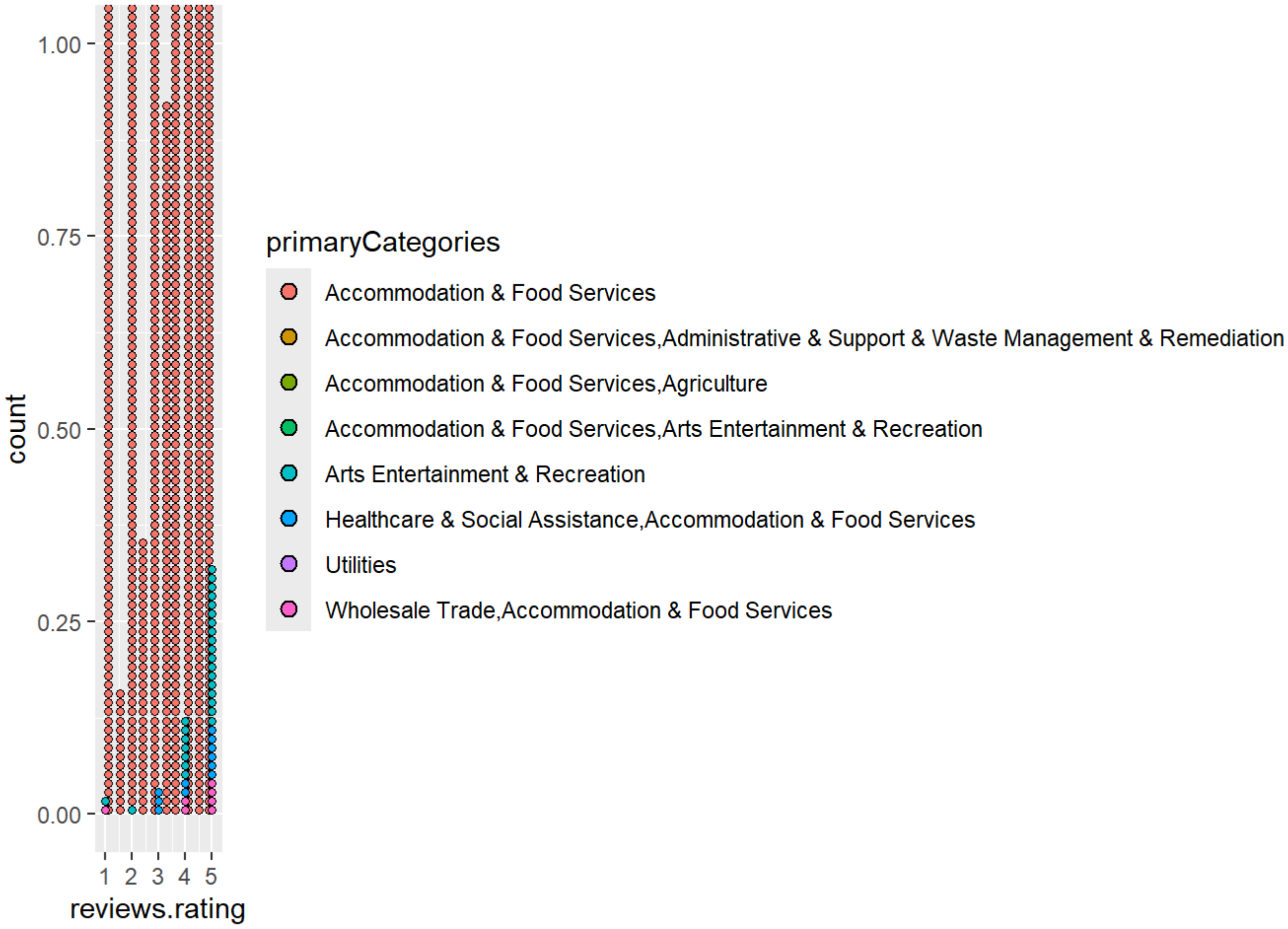
```
Bin width defaults to 1/30 of the range of the data. Pick better value with
`binwidth`.

Warning: Removed 862 rows containing missing values or values outside the scale range
(`stat_bindot()`).
```




```
# Shows dot plot grouped by category
ggplot(df, aes(x = reviews.rating, fill = primaryCategories)) + geom_dotplot()
```

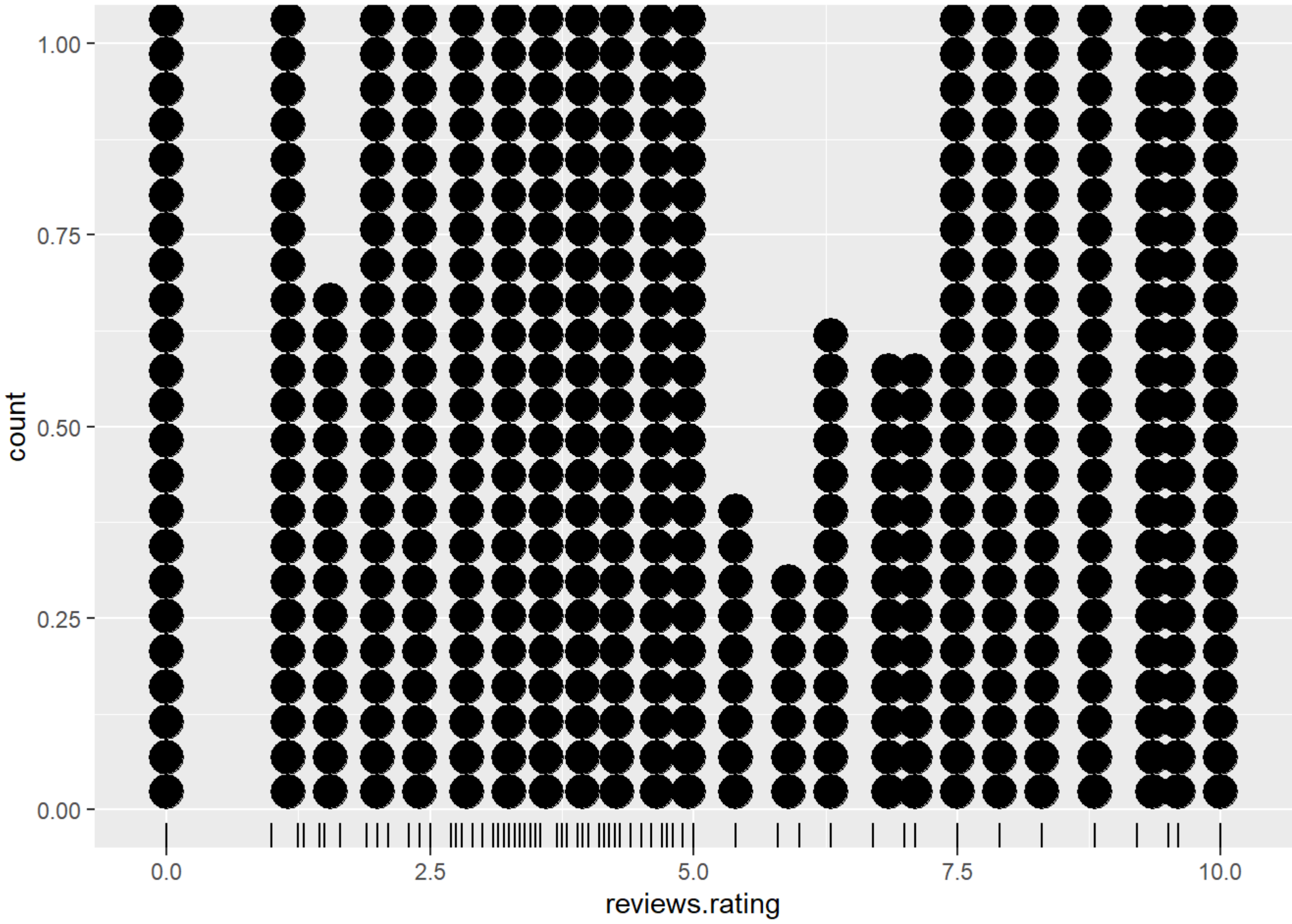
```
Bin width defaults to 1/30 of the range of the data. Pick better value with
`binwidth`.
```

```
Warning: Removed 35912 rows containing missing values or values outside the scale range
(`stat_bindot()`).
```




```
# Shows dot plot with rug marks
ggplot(df, aes(x = reviews.rating)) +
  geom_dotplot() + geom_rug()
```

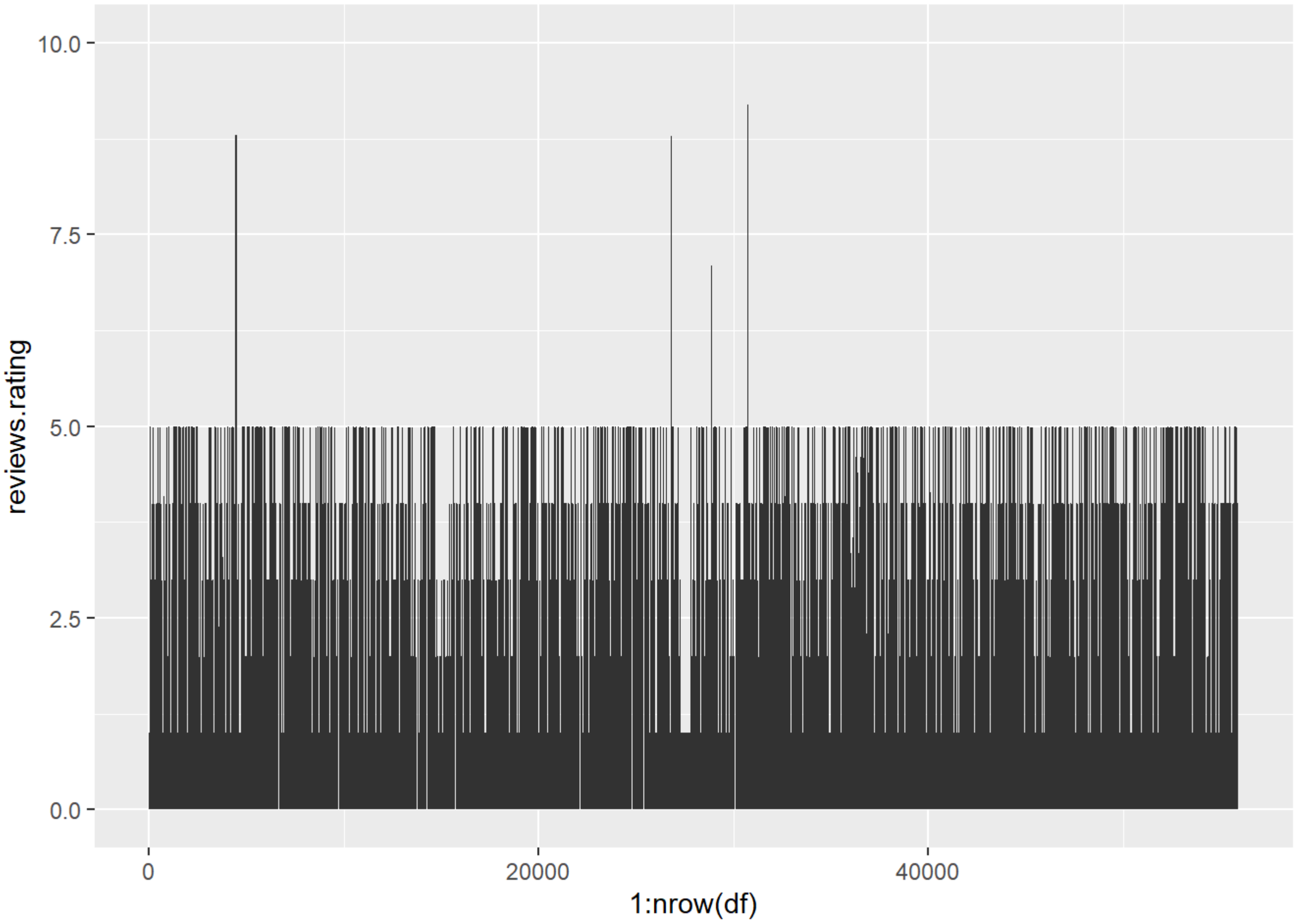
```
Bin width defaults to 1/30 of the range of the data. Pick better value with
`binwidth`.

Warning: Removed 862 rows containing missing values or values outside the scale range
(`stat_bindot()`).

Warning: Removed 862 rows containing missing values or values outside the scale range
(`geom_rug()`).
```



AREA / TREND

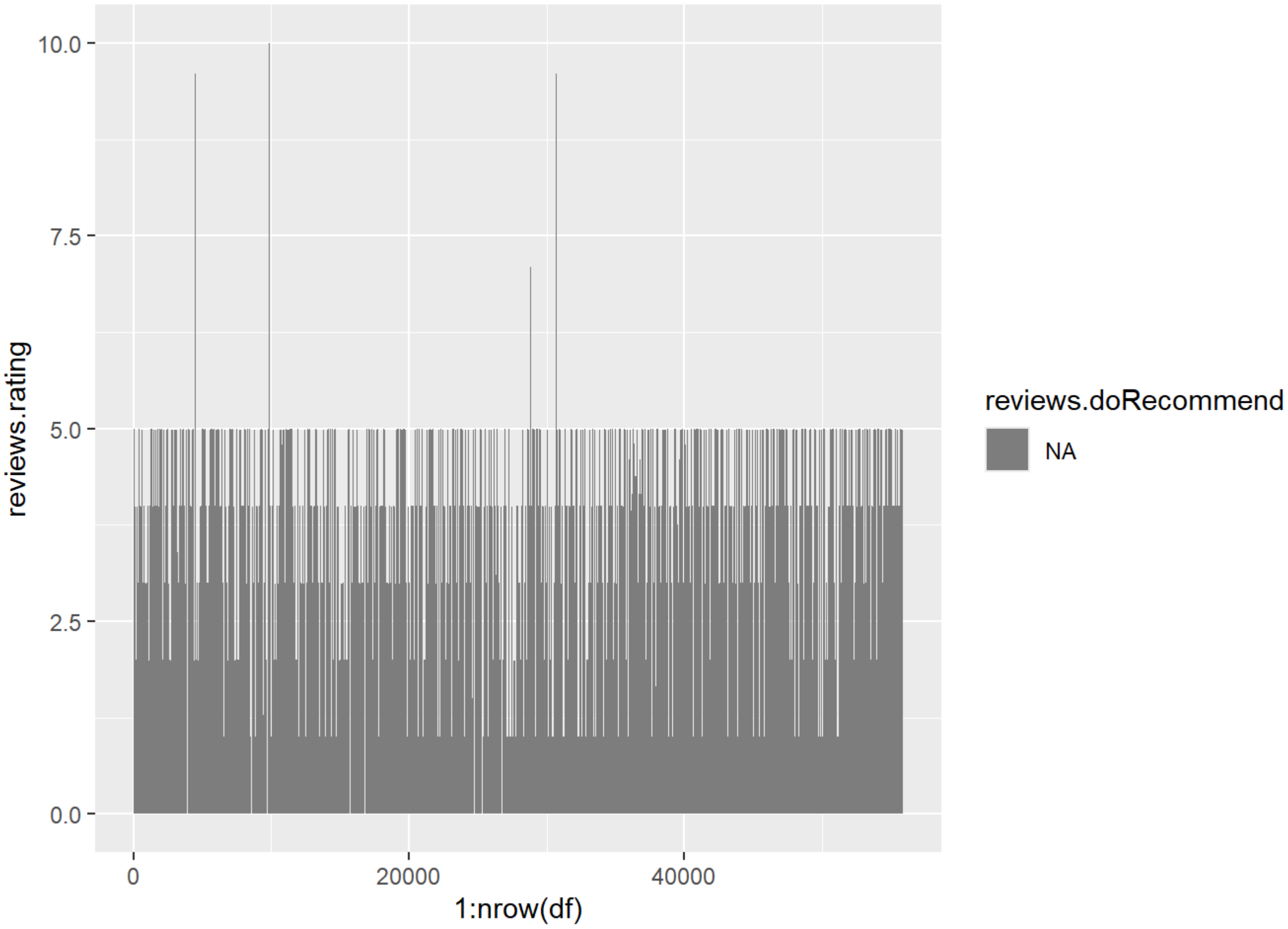
```
# Shows area representing rating trend
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating)) + geom_area()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_align()`).
```



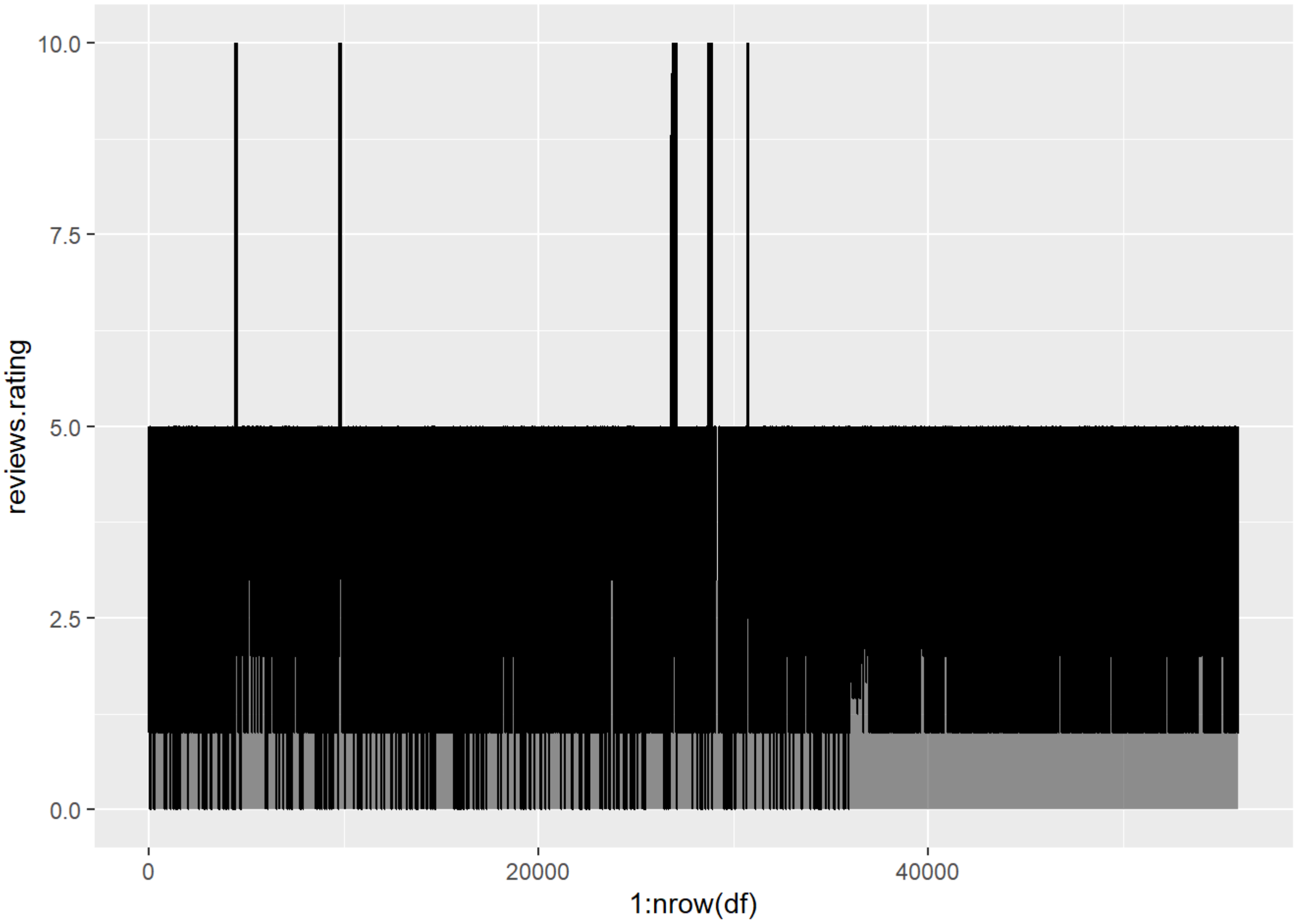

```
# Shows area grouped by recommendation
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating, fill = reviews.doRecommend)) + geom_area()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_align()`).
```



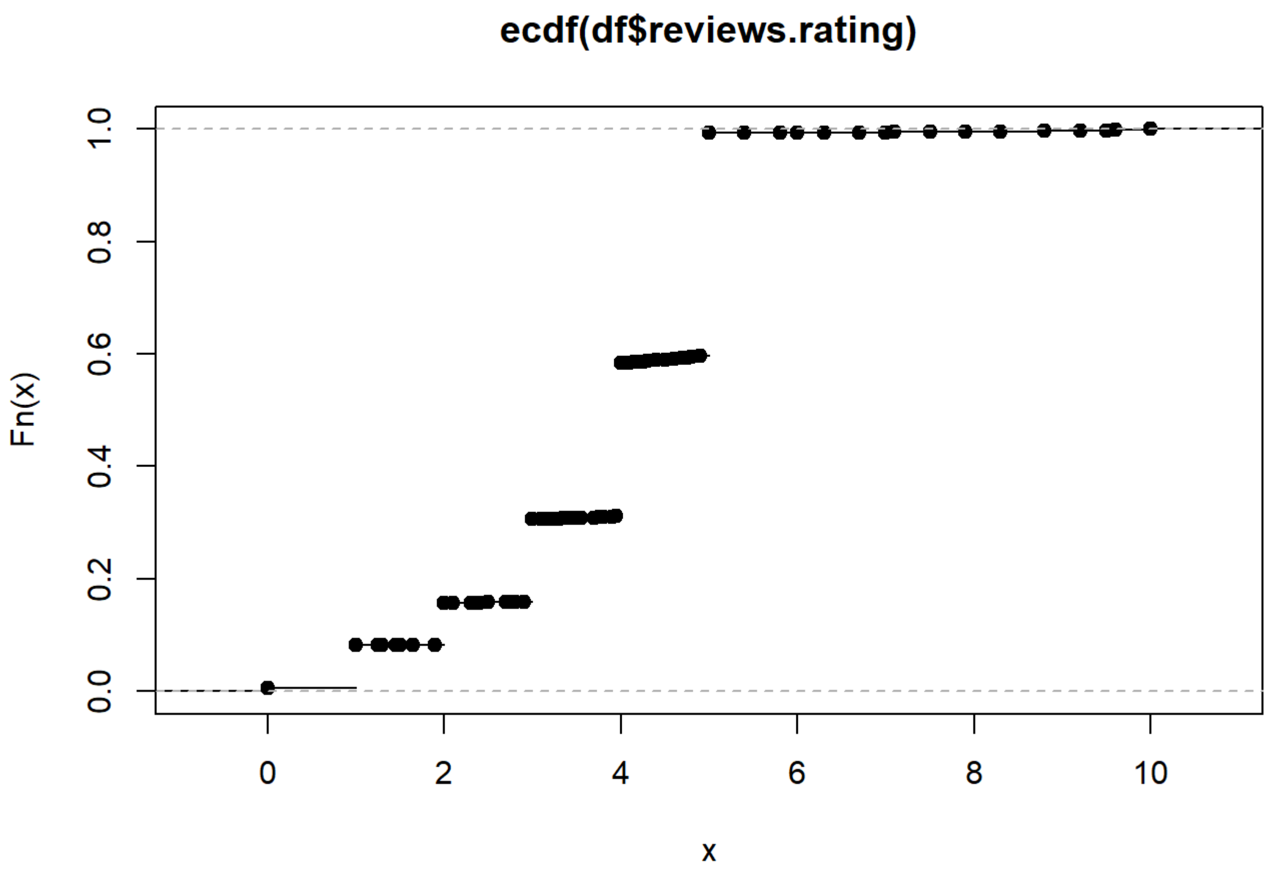

```r
# Shows combined area and line trend
ggplot(df, aes(x = 1:nrow(df), y = reviews.rating)) +
  geom_area(alpha=0.5) + geom_line()
```

```
Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_align()`).
```



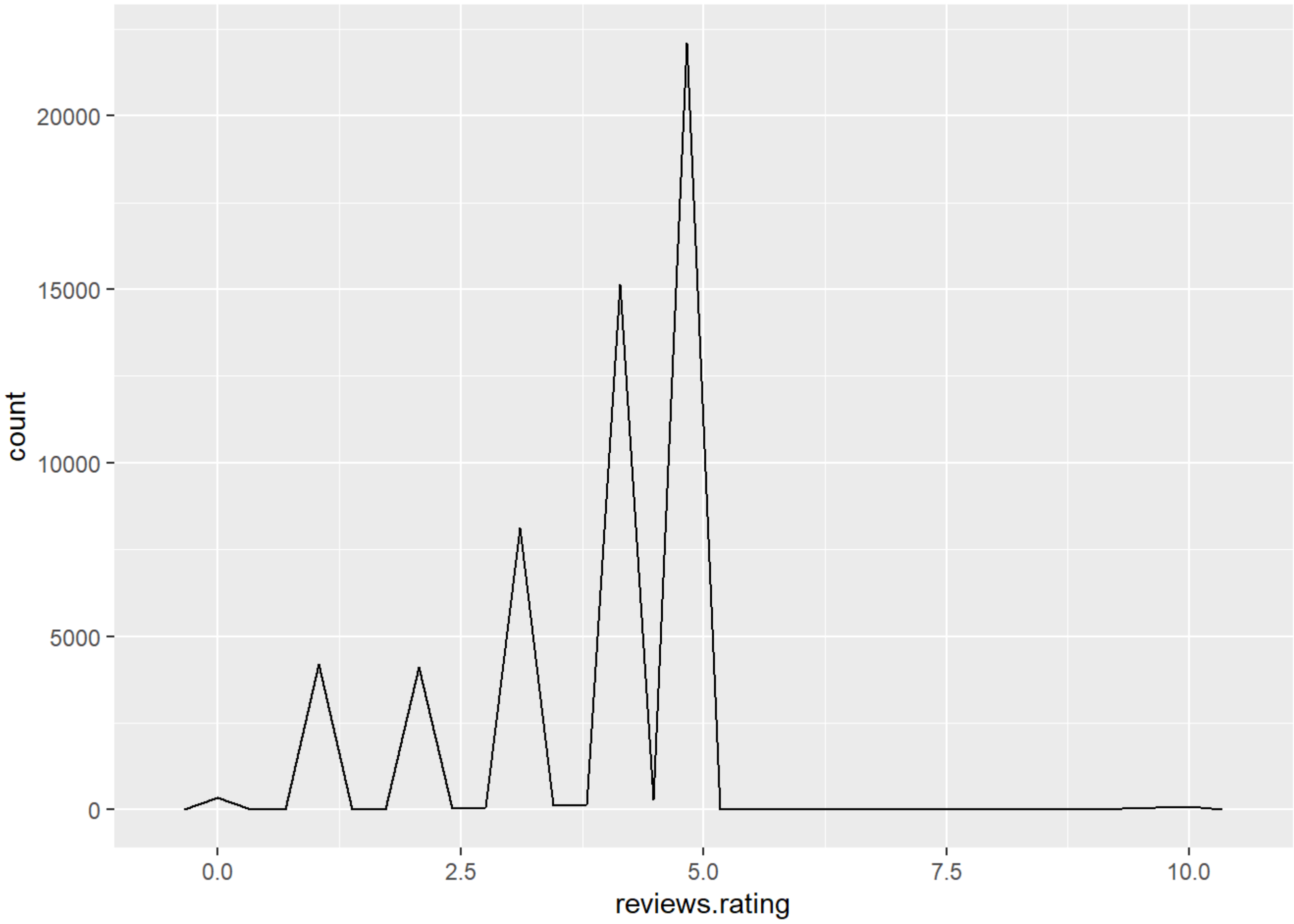
SPECIAL PLOTS

```
# Shows cumulative distribution of ratings
plot(ecdf(df$reviews.rating))
```

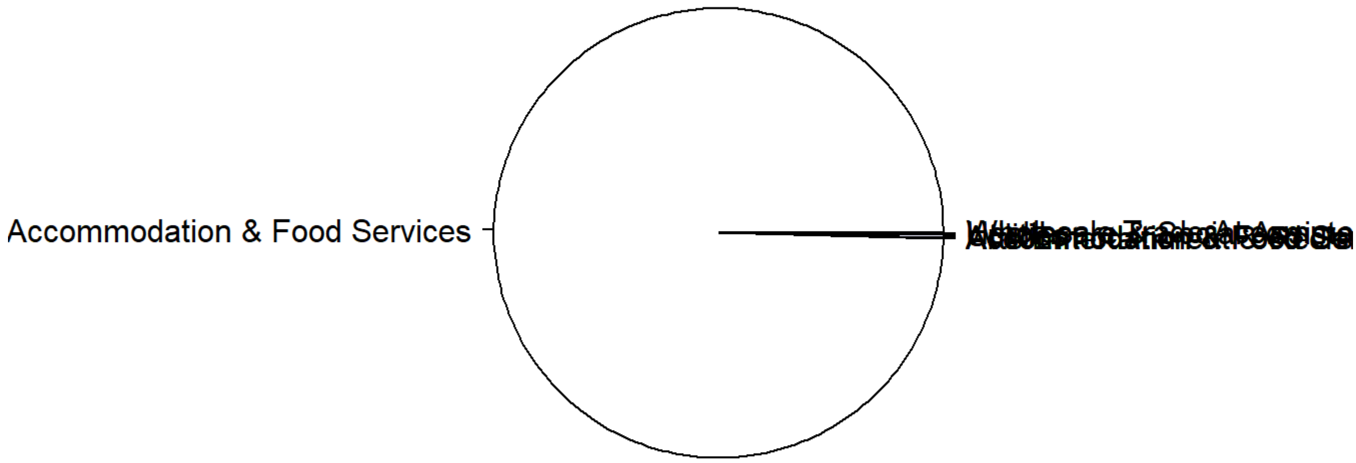



```r
# Shows frequency distribution as polygon
ggplot(df, aes(x = reviews.rating)) + geom_freqpoly()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 862 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
# Shows proportion of categories using pie chart
pie(table(df$primaryCategories))
```


Accommodation & Food Services